

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12403315
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Jayakumar; Adarsh et al.

Systems and methods for automated programming of electrical stimulation

Abstract

A method for automating selection of stimulation parameters for a stimulation device implanted in a patient includes setting, by a user, at least one limit on each of at least one stimulation parameter and performing, automatically using at least one processor, the following actions for each of a plurality of sets of the stimulation parameters constrained by the at least one limit: stimulating the patient, by the stimulation device, using the set of stimulation parameters, sensing one or more effects arising in response to the stimulation, and updating, by the at least one processor, a collection of the effects and sets of stimulation parameters with the one or more effects and the set of stimulation parameters. The method further includes selecting, by the processor, one of the sets of stimulation parameters based on the effects.

Inventors: Jayakumar; Adarsh (Valencia, CA), Doan; Thien Tich (West Hills, CA)

Applicant: Boston Scientific Neuromodulation Corporation (Valencia, CA)

Family ID: 81654755

Assignee: Boston Scientific Neuromodulation Corporation (Valencia, CA)

Appl. No.: 17/728493

Filed: April 25, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220339448 A1	Oct. 27, 2022

Related U.S. Application Data

us-provisional-application US 63180380 20210427

Publication Classification

Int. Cl.: A61N1/36 (20060101); A61N1/372 (20060101)

U.S. Cl.:

CPC A61N1/36139 (20130101); A61N1/37247 (20130101);

Field of Classification Search

CPC: A61N (1/36139); A61N (1/37247); A61N (1/36); A61N (1/36146); A61N (1/36135)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3999555	12/1975	Person	N/A	N/A
4144889	12/1978	Tyers et al.	N/A	N/A
4177818	12/1978	De Pedro	N/A	N/A
4341221	12/1981	Testerman	N/A	N/A
4378797	12/1982	Osterholm	N/A	N/A
4445500	12/1983	Osterholm	N/A	N/A
4735208	12/1987	Wyler et al.	N/A	N/A
4765341	12/1987	Mower et al.	N/A	N/A
4841973	12/1988	Stecker	N/A	N/A
5067495	12/1990	Brehm	N/A	N/A
5099846	12/1991	Hardy	N/A	N/A
5222494	12/1992	Baker, Jr.	N/A	N/A
5255693	12/1992	Dutcher	N/A	N/A
5259387	12/1992	dePinto	N/A	N/A
5304206	12/1993	Baker, Jr. et al.	N/A	N/A
5344438	12/1993	Testerman et al.	N/A	N/A
5361763	12/1993	Kao et al.	N/A	N/A
5452407	12/1994	Crook	N/A	N/A
5560360	12/1995	Filler et al.	N/A	N/A
5565949	12/1995	Kasha, Jr.	N/A	N/A
5593427	12/1996	Gliner et al.	N/A	N/A
5601612	12/1996	Gliner et al.	N/A	N/A
5607454	12/1996	Cameron et al.	N/A	N/A
5620470	12/1996	Gliner et al.	N/A	N/A
5651767	12/1996	Schulman	N/A	N/A
5711316	12/1997	Elsberry et al.	N/A	N/A
5713922	12/1997	King	N/A	N/A
5716377	12/1997	Rise et al.	N/A	N/A
5724985	12/1997	Snell et al.	N/A	N/A
5749904	12/1997	Gliner et al.	N/A	N/A
5749905	12/1997	Gliner et al.	N/A	N/A
5776170	12/1997	MacDonald et al.	N/A	N/A
5782762	12/1997	Vining	N/A	N/A

5843148	12/1997	Gijsbers et al.	N/A	N/A
5859922	12/1998	Hoffmann	N/A	N/A
5868740	12/1998	LeVeen et al.	N/A	N/A
5897583	12/1998	Meyer et al.	N/A	N/A
5910804	12/1998	Fortenbery et al.	N/A	N/A
5925070	12/1998	King et al.	N/A	N/A
5938688	12/1998	Schiff	N/A	N/A
5938690	12/1998	Law et al.	N/A	N/A
5978713	12/1998	Prutchi et al.	N/A	N/A
6016449	12/1999	Fischell et al.	N/A	N/A
6029090	12/1999	Herbst	N/A	N/A
6029091	12/1999	de la Rama et al.	N/A	N/A
6050992	12/1999	Nichols	N/A	N/A
6058331	12/1999	King	N/A	N/A
6066163	12/1999	John	N/A	N/A
6080187	12/1999	Alt et al.	N/A	N/A
6083162	12/1999	Vining	N/A	N/A
6094598	12/1999	Elsberry et al.	N/A	N/A
6096756	12/1999	Crain et al.	N/A	N/A
6106460	12/1999	Panescu et al.	N/A	N/A
6109269	12/1999	Rise et al.	N/A	N/A
6128538	12/1999	Fischell et al.	N/A	N/A
6129685	12/1999	Howard, III	N/A	N/A
6146390	12/1999	Heilbrun et al.	N/A	N/A
6161044	12/1999	Silverstone	N/A	N/A
6167311	12/1999	Rezai	N/A	N/A
6181969	12/2000	Gord	N/A	N/A
6192266	12/2000	Dupree et al.	N/A	N/A
6205361	12/2000	Kuzma	N/A	N/A
6208881	12/2000	Champeau	N/A	N/A
6240308	12/2000	Hardy et al.	N/A	N/A
6246912	12/2000	Sluijter et al.	N/A	N/A
6253109	12/2000	Gielen	N/A	N/A
6289239	12/2000	Panescu et al.	N/A	N/A
6301492	12/2000	Zonenshayn	N/A	N/A
6310619	12/2000	Rice	N/A	N/A
6319241	12/2000	King	N/A	N/A
6336899	12/2001	Yamazaki	N/A	N/A
6343226	12/2001	Sunde et al.	N/A	N/A
6351675	12/2001	Tholen et al.	N/A	N/A
6353762	12/2001	Baudino et al.	N/A	N/A
6366813	12/2001	Dilorenzo	N/A	N/A
6368331	12/2001	Front et al.	N/A	N/A
6389311	12/2001	Whayne et al.	N/A	N/A
6393325	12/2001	Mann et al.	N/A	N/A
6421566	12/2001	Holsheimer	N/A	N/A
6435878	12/2001	Reynolds et al.	N/A	N/A
6442432	12/2001	Lee	N/A	N/A
6463328	12/2001	John	N/A	N/A
6491699	12/2001	Henderson et al.	N/A	N/A

6494831	12/2001	Koritzinsky	N/A	N/A
6507759	12/2002	Prutchi et al.	N/A	N/A
6510347	12/2002	Borkan	N/A	N/A
6516227	12/2002	Meadows et al.	N/A	N/A
6517480	12/2002	Krass	N/A	N/A
6539263	12/2002	Schiff	N/A	N/A
6560490	12/2002	Grill et al.	N/A	N/A
6579280	12/2002	Kovach et al.	N/A	N/A
6600956	12/2002	Maschino et al.	N/A	N/A
6606523	12/2002	Jenkins	N/A	N/A
6609029	12/2002	Mann et al.	N/A	N/A
6609031	12/2002	Law et al.	N/A	N/A
6609032	12/2002	Woods et al.	N/A	N/A
6622048	12/2002	Mann et al.	N/A	N/A
6631297	12/2002	Mo	N/A	N/A
6654642	12/2002	North et al.	N/A	N/A
6662053	12/2002	Borkan	N/A	N/A
6675046	12/2003	Holsheimer	N/A	N/A
6684106	12/2003	Herbst	N/A	N/A
6687392	12/2003	Touzawa et al.	N/A	N/A
6690972	12/2003	Conley et al.	N/A	N/A
6690974	12/2003	Archer et al.	N/A	N/A
6692315	12/2003	Soumillion et al.	N/A	N/A
6694162	12/2003	Hartlep	N/A	N/A
6694163	12/2003	Vining	N/A	N/A
6708096	12/2003	Frei et al.	N/A	N/A
6741892	12/2003	Meadows et al.	N/A	N/A
6748098	12/2003	Rosenfeld	N/A	N/A
6748276	12/2003	Daignault, Jr. et al.	N/A	N/A
6778846	12/2003	Martinez et al.	N/A	N/A
6788969	12/2003	Dupree et al.	N/A	N/A
6795737	12/2003	Gielen et al.	N/A	N/A
6827681	12/2003	Tanner et al.	N/A	N/A
6830544	12/2003	Tanner	N/A	N/A
6845267	12/2004	Harrison et al.	N/A	N/A
6850802	12/2004	Holsheimer	N/A	N/A
6895280	12/2004	Meadows et al.	N/A	N/A
6909913	12/2004	Vining	N/A	N/A
6937891	12/2004	Leinders et al.	N/A	N/A
6937903	12/2004	Schuler et al.	N/A	N/A
6944497	12/2004	Stypulkowski	N/A	N/A
6944501	12/2004	Pless	N/A	N/A
6950707	12/2004	Whitehurst	N/A	N/A
6969388	12/2004	Goldman et al.	N/A	N/A
6993384	12/2005	Bradley et al.	N/A	N/A
7003349	12/2005	Andersson et al.	N/A	N/A
7003352	12/2005	Whitehurst	N/A	N/A
7008370	12/2005	Tanner et al.	N/A	N/A
7008413	12/2005	Kovach et al.	N/A	N/A
7035690	12/2005	Goetz	N/A	N/A

7043293	12/2005	Baura	N/A	N/A
7047082	12/2005	Schrom et al.	N/A	N/A
7047084	12/2005	Erickson et al.	N/A	N/A
7050857	12/2005	Samuelsson et al.	N/A	N/A
7054692	12/2005	Whitehurst et al.	N/A	N/A
7058446	12/2005	Schuler et al.	N/A	N/A
7082333	12/2005	Bauhahn et al.	N/A	N/A
7107102	12/2005	Daignault et al.	N/A	N/A
7126000	12/2005	Ogawa et al.	N/A	N/A
7127297	12/2005	Law et al.	N/A	N/A
7136518	12/2005	Griffin et al.	N/A	N/A
7136695	12/2005	Pless et al.	N/A	N/A
7142923	12/2005	North et al.	N/A	N/A
7146219	12/2005	Sieracki et al.	N/A	N/A
7146223	12/2005	King	N/A	N/A
7151961	12/2005	Whitehurst	N/A	N/A
7155279	12/2005	Whitehurst	N/A	N/A
7167760	12/2006	Dawant et al.	N/A	N/A
7177674	12/2006	Echauz et al.	N/A	N/A
7181286	12/2006	Sieracki et al.	N/A	N/A
7184837	12/2006	Goetz	N/A	N/A
7191014	12/2006	Kobayashi et al.	N/A	N/A
7209787	12/2006	Dilorenzo	N/A	N/A
7211050	12/2006	Caplygin	N/A	N/A
7216000	12/2006	Sieracki et al.	N/A	N/A
7217276	12/2006	Henderson	N/A	N/A
7218968	12/2006	Condie et al.	N/A	N/A
7228179	12/2006	Campen et al.	N/A	N/A
7231254	12/2006	DiLorenzo	N/A	N/A
7236830	12/2006	Gliner	N/A	N/A
7239910	12/2006	Tanner	N/A	N/A
7239916	12/2006	Thompson et al.	N/A	N/A
7239926	12/2006	Goetz	N/A	N/A
7242984	12/2006	DiLorenzo	N/A	N/A
7244150	12/2006	Brase et al.	N/A	N/A
7252090	12/2006	Goetz	N/A	N/A
7254445	12/2006	Law et al.	N/A	N/A
7254446	12/2006	Erickson	N/A	N/A
7257447	12/2006	Cates et al.	N/A	N/A
7266412	12/2006	Stypulkowski	N/A	N/A
7289761	12/2006	Mazar et al.	N/A	N/A
7294107	12/2006	Simon et al.	N/A	N/A
7295876	12/2006	Erickson	N/A	N/A
7299096	12/2006	Balzer et al.	N/A	N/A
7308302	12/2006	Schuler et al.	N/A	N/A
7313430	12/2006	Urquhart	N/A	N/A
7324851	12/2007	DiLorenzo	N/A	N/A
7346282	12/2007	Sakanaka et al.	N/A	N/A
7346382	12/2007	McIntyre et al.	N/A	N/A
7385443	12/2007	Denison	N/A	N/A

7388974	12/2007	Yanagita	N/A	N/A
7437193	12/2007	Parramon et al.	N/A	N/A
7450997	12/2007	Pianca et al.	N/A	N/A
7454245	12/2007	Armstrong et al.	N/A	N/A
7463928	12/2007	Lee et al.	N/A	N/A
7499048	12/2008	Sieracki et al.	N/A	N/A
7505815	12/2008	Lee et al.	N/A	N/A
7548786	12/2008	Lee et al.	N/A	N/A
7565199	12/2008	Sheffield et al.	N/A	N/A
7603177	12/2008	Sieracki et al.	N/A	N/A
7617002	12/2008	Goetz	N/A	N/A
7623918	12/2008	Goetz	N/A	N/A
7650184	12/2009	Walter	N/A	N/A
7657319	12/2009	Goetz et al.	N/A	N/A
7672734	12/2009	Anderson et al.	N/A	N/A
7676273	12/2009	Goetz et al.	N/A	N/A
7680526	12/2009	McIntyre et al.	N/A	N/A
7734340	12/2009	De Ridder	N/A	N/A
7761165	12/2009	He et al.	N/A	N/A
7783359	12/2009	Meadows	N/A	N/A
7792590	12/2009	Pianca et al.	N/A	N/A
7809446	12/2009	Meadows	N/A	N/A
7826902	12/2009	Stone et al.	N/A	N/A
7848802	12/2009	Goetz et al.	N/A	N/A
7860548	12/2009	McIntyre et al.	N/A	N/A
7896808	12/2010	Koh et al.	N/A	N/A
7904134	12/2010	McIntyre et al.	N/A	N/A
7945105	12/2010	Jaenisch	N/A	N/A
7949395	12/2010	Kuzma	N/A	N/A
7974706	12/2010	Moffitt et al.	N/A	N/A
8000794	12/2010	Lozano	N/A	N/A
8019439	12/2010	Kuzma et al.	N/A	N/A
8019443	12/2010	Schleicher et al.	N/A	N/A
8175710	12/2011	He	N/A	N/A
8180601	12/2011	Butson et al.	N/A	N/A
8187209	12/2011	Giuffrida	N/A	N/A
8195300	12/2011	Gliner et al.	N/A	N/A
8209027	12/2011	Butson et al.	N/A	N/A
8224450	12/2011	Brase	N/A	N/A
8257684	12/2011	Covalin et al.	N/A	N/A
8262714	12/2011	Hulvershorn et al.	N/A	N/A
8271094	12/2011	Moffitt et al.	N/A	N/A
8280514	12/2011	Lozano et al.	N/A	N/A
8295944	12/2011	Howard et al.	N/A	N/A
8306627	12/2011	Armstrong	N/A	N/A
8326433	12/2011	Blum et al.	N/A	N/A
8359107	12/2012	Pianca et al.	N/A	N/A
8364278	12/2012	Pianca et al.	N/A	N/A
8369954	12/2012	Stack et al.	N/A	N/A
8379952	12/2012	McIntyre et al.	N/A	N/A

8391985	12/2012	McDonald	N/A	N/A
8412349	12/2012	Barker	N/A	N/A
8429174	12/2012	Ramani et al.	N/A	N/A
8452415	12/2012	Goetz et al.	N/A	N/A
8467883	12/2012	Chen et al.	N/A	N/A
8473061	12/2012	Moffitt et al.	N/A	N/A
8483237	12/2012	Zimmermann et al.	N/A	N/A
8543189	12/2012	Paitel et al.	N/A	N/A
8571665	12/2012	Moffitt et al.	N/A	N/A
8589316	12/2012	Lujan et al.	N/A	N/A
8594800	12/2012	Butson et al.	N/A	N/A
8594801	12/2012	Corndorf et al.	N/A	N/A
8606360	12/2012	Butson et al.	N/A	N/A
8620452	12/2012	King et al.	N/A	N/A
8649845	12/2013	McIntyre et al.	N/A	N/A
8675945	12/2013	Barnhorst et al.	N/A	N/A
8679038	12/2013	Giuffrida	N/A	N/A
8688235	12/2013	Pianca et al.	N/A	N/A
8744596	12/2013	Howard	N/A	N/A
8751008	12/2013	Carlton et al.	N/A	N/A
8751016	12/2013	Schleicher et al.	N/A	N/A
8774941	12/2013	Pianca	N/A	N/A
8792993	12/2013	Pianca et al.	N/A	N/A
8831731	12/2013	Blum et al.	N/A	N/A
8831742	12/2013	Pianca et al.	N/A	N/A
8845557	12/2013	Giuffrida et al.	N/A	N/A
8849632	12/2013	Sparks et al.	N/A	N/A
8855773	12/2013	Kokones et al.	N/A	N/A
8868199	12/2013	Kaula et al.	N/A	N/A
8913804	12/2013	Blum et al.	N/A	N/A
8918183	12/2013	Carlton et al.	N/A	N/A
8918184	12/2013	Torgerson et al.	N/A	N/A
8923976	12/2013	Johanek	N/A	N/A
8936622	12/2014	Wales et al.	N/A	N/A
8958615	12/2014	Blum et al.	N/A	N/A
8972023	12/2014	Bradley et al.	N/A	N/A
8986382	12/2014	Bentley et al.	N/A	N/A
9020789	12/2014	Butson et al.	N/A	N/A
9026317	12/2014	Furukawa et al.	N/A	N/A
9039740	12/2014	Wales et al.	N/A	N/A
9050470	12/2014	Carlton et al.	N/A	N/A
9061138	12/2014	Pianca	N/A	N/A
9072905	12/2014	Kokones et al.	N/A	N/A
9081488	12/2014	Soederstroem	N/A	N/A
9084896	12/2014	Kokones et al.	N/A	N/A
9135400	12/2014	McIntyre et al.	N/A	N/A
9149630	12/2014	Howard et al.	N/A	N/A
9162056	12/2014	Pianca	N/A	N/A
9220889	12/2014	Carlton et al.	N/A	N/A
9227074	12/2015	Carcieri et al.	N/A	N/A

9235685	12/2015	McIntyre et al.	N/A	N/A
9248272	12/2015	Romero	N/A	N/A
9248296	12/2015	Carcieri et al.	N/A	N/A
9254387	12/2015	Blum et al.	N/A	N/A
9272153	12/2015	Blum et al.	N/A	N/A
9289596	12/2015	Leven	N/A	N/A
9289600	12/2015	Govea et al.	N/A	N/A
9302110	12/2015	Kokones et al.	N/A	N/A
9308372	12/2015	Sparks et al.	N/A	N/A
9310985	12/2015	Blum et al.	N/A	N/A
9327111	12/2015	Pianca et al.	N/A	N/A
9358398	12/2015	Moffitt et al.	N/A	N/A
9364665	12/2015	Bokil et al.	N/A	N/A
9381348	12/2015	Romero et al.	N/A	N/A
9387325	12/2015	Min et al.	N/A	N/A
9415154	12/2015	Leven	N/A	N/A
9474903	12/2015	Chen et al.	N/A	N/A
9492655	12/2015	Pianca et al.	N/A	N/A
9498620	12/2015	Romero et al.	N/A	N/A
9526898	12/2015	Libbus	N/A	A61N 1/0551
9526902	12/2015	Blum et al.	N/A	N/A
9533141	12/2016	Black et al.	N/A	N/A
9566596	12/2016	Kim et al.	N/A	N/A
9572982	12/2016	Burnes et al.	N/A	N/A
9586053	12/2016	Moffitt et al.	N/A	N/A
9592389	12/2016	Moffitt	N/A	N/A
9610435	12/2016	Schleicher et al.	N/A	N/A
9636498	12/2016	Leven	N/A	N/A
9649489	12/2016	Wechter et al.	N/A	N/A
9669210	12/2016	Barker et al.	N/A	N/A
9713720	12/2016	Zhu	N/A	N/A
9775988	12/2016	Govea et al.	N/A	N/A
9792412	12/2016	Moffitt et al.	N/A	N/A
9821167	12/2016	Carcieri et al.	N/A	N/A
9887470	12/2017	Nguyen-Stella et al.	N/A	N/A
9925382	12/2017	Carlton et al.	N/A	N/A
9959940	12/2017	Moffitt et al.	N/A	N/A
9974959	12/2017	Moffitt et al.	N/A	N/A
9987482	12/2017	Nageri et al.	N/A	N/A
10067659	12/2017	Bokil	N/A	N/A
10071242	12/2017	Leven	N/A	N/A
10071249	12/2017	Zottola	N/A	N/A
10086205	12/2017	Grill et al.	N/A	N/A
10213148	12/2018	Min et al.	N/A	N/A
10226616	12/2018	Barker	N/A	N/A
10265528	12/2018	Carcieri et al.	N/A	N/A
10265531	12/2018	Bokil	N/A	N/A
10286205	12/2018	Steinke et al.	N/A	N/A
10300282	12/2018	Torgerson et al.	N/A	N/A

10335607	12/2018	Orinski	N/A	N/A
10357657	12/2018	Moffitt et al.	N/A	N/A
10369364	12/2018	Moffitt et al.	N/A	N/A
10406353	12/2018	Wechter	N/A	N/A
10485969	12/2018	Govea et al.	N/A	N/A
10493269	12/2018	Stoffregen et al.	N/A	N/A
10525257	12/2019	Govea et al.	N/A	N/A
10525266	12/2019	Moffitt et al.	N/A	N/A
10603498	12/2019	Blum et al.	N/A	N/A
10625072	12/2019	Serrano Carmona	N/A	N/A
10631937	12/2019	Tyulmankov et al.	N/A	N/A
10639488	12/2019	Kalgren et al.	N/A	N/A
10653330	12/2019	Angle et al.	N/A	N/A
10675468	12/2019	Torgerson	N/A	N/A
10709886	12/2019	Nagaoka et al.	N/A	N/A
10716505	12/2019	Blum et al.	N/A	N/A
10780282	12/2019	Mustakos et al.	N/A	N/A
10814127	12/2019	Nageri et al.	N/A	N/A
10814140	12/2019	Zhang et al.	N/A	N/A
10835739	12/2019	Sandhu	N/A	N/A
10850101	12/2019	Zhang et al.	N/A	N/A
10857351	12/2019	Wang et al.	N/A	N/A
10960203	12/2020	Tyler et al.	N/A	N/A
11020052	12/2020	Zuckerman-Stark et al.	N/A	N/A
11285329	12/2021	Carcieri et al.	N/A	N/A
11298550	12/2021	Howard et al.	N/A	N/A
11357986	12/2021	Steinke et al.	N/A	N/A
11517755	12/2021	Zhang et al.	N/A	N/A
11529510	12/2021	Leven	N/A	N/A
11707622	12/2022	Juarez Paz et al.	N/A	N/A
11745010	12/2022	Donega et al.	N/A	N/A
2001/0031071	12/2000	Nichols et al.	N/A	N/A
2002/0032375	12/2001	Bauch et al.	N/A	N/A
2002/0062143	12/2001	Baudino et al.	N/A	N/A
2002/0087201	12/2001	Firlik et al.	N/A	N/A
2002/0099295	12/2001	Gil et al.	N/A	N/A
2002/0115603	12/2001	Whitehouse	N/A	N/A
2002/0116030	12/2001	Rezei	N/A	N/A
2002/0123780	12/2001	Grill et al.	N/A	N/A
2002/0128694	12/2001	Holsheimer	N/A	N/A
2002/0151939	12/2001	Rezai	N/A	N/A
2002/0183607	12/2001	Bauch et al.	N/A	N/A
2002/0183740	12/2001	Edwards et al.	N/A	N/A
2002/0183817	12/2001	Van Venrooij et al.	N/A	N/A
2003/0097159	12/2002	Schiff et al.	N/A	N/A
2003/0149450	12/2002	Mayberg	N/A	N/A
2003/0171791	12/2002	KenKnight et al.	N/A	N/A
2003/0212439	12/2002	Schuler et al.	N/A	N/A
2004/0034394	12/2003	Woods et al.	N/A	N/A
2004/0044279	12/2003	Lewin et al.	N/A	N/A

2004/0044378	12/2003	Holsheimer	N/A	N/A
2004/0044379	12/2003	Holsheimer	N/A	N/A
2004/0054297	12/2003	Wingeier	600/544	A61B 5/165
2004/0059395	12/2003	North et al.	N/A	N/A
2004/0106916	12/2003	Quaid et al.	N/A	N/A
2004/0133248	12/2003	Frei et al.	N/A	N/A
2004/0152957	12/2003	Stivoric et al.	N/A	N/A
2004/0181262	12/2003	Bauhahn	N/A	N/A
2004/0186532	12/2003	Tadlock	N/A	N/A
2004/0193231	12/2003	David et al.	N/A	N/A
2004/0199216	12/2003	Lee et al.	N/A	N/A
2004/0267330	12/2003	Lee et al.	N/A	N/A
2005/0021090	12/2004	Schuler et al.	N/A	N/A
2005/0033380	12/2004	Tanner et al.	N/A	N/A
2005/0049649	12/2004	Luders et al.	N/A	N/A
2005/0060001	12/2004	Singhal et al.	N/A	N/A
2005/0060009	12/2004	Goetz	N/A	N/A
2005/0070781	12/2004	Dawant et al.	N/A	N/A
2005/0075689	12/2004	Toy et al.	N/A	N/A
2005/0085714	12/2004	Foley et al.	N/A	N/A
2005/0165294	12/2004	Weiss	N/A	N/A
2005/0171587	12/2004	Daglow et al.	N/A	N/A
2005/0228250	12/2004	Bitter et al.	N/A	N/A
2005/0251061	12/2004	Schuler et al.	N/A	N/A
2005/0261061	12/2004	Nguyen et al.	N/A	N/A
2005/0261601	12/2004	Schuler et al.	N/A	N/A
2005/0261747	12/2004	Schuler et al.	N/A	N/A
2005/0267347	12/2004	Oster	N/A	N/A
2005/0288732	12/2004	Schuler et al.	N/A	N/A
2006/0004422	12/2005	De Ridder	N/A	N/A
2006/0017749	12/2005	McIntyre et al.	N/A	N/A
2006/0020292	12/2005	Goetz et al.	N/A	N/A
2006/0069415	12/2005	Cameron et al.	N/A	N/A
2006/0094951	12/2005	Dean et al.	N/A	N/A
2006/0095088	12/2005	Riddler	N/A	N/A
2006/0155340	12/2005	Schuler et al.	N/A	N/A
2006/0206169	12/2005	Schuler	N/A	N/A
2006/0218007	12/2005	Bjorner et al.	N/A	N/A
2006/0224189	12/2005	Schuler et al.	N/A	N/A
2006/0235472	12/2005	Goetz et al.	N/A	N/A
2006/0259079	12/2005	King	N/A	N/A
2006/0259099	12/2005	Goetz et al.	N/A	N/A
2007/0000372	12/2006	Rezai et al.	N/A	N/A
2007/0017749	12/2006	Dold et al.	N/A	N/A
2007/0027499	12/2006	Maschino et al.	N/A	N/A
2007/0027514	12/2006	Gerber	N/A	N/A
2007/0043268	12/2006	Russell	N/A	N/A
2007/0049817	12/2006	Preiss et al.	N/A	N/A
2007/0067003	12/2006	Sanchez et al.	N/A	N/A
2007/0078498	12/2006	Rezai et al.	N/A	N/A

2007/0083104	12/2006	Butson et al.	N/A	N/A
2007/0123953	12/2006	Lee et al.	N/A	N/A
2007/0129769	12/2006	Bourget et al.	N/A	N/A
2007/0135855	12/2006	Foshee et al.	N/A	N/A
2007/0150026	12/2006	Bourget et al.	N/A	N/A
2007/0150036	12/2006	Anderson	N/A	N/A
2007/0156186	12/2006	Lee et al.	N/A	N/A
2007/0162086	12/2006	DiLorenzo	N/A	N/A
2007/0162235	12/2006	Zhan et al.	N/A	N/A
2007/0168004	12/2006	Walter	N/A	N/A
2007/0168007	12/2006	Kuzma et al.	N/A	N/A
2007/0179557	12/2006	Maschino et al.	N/A	N/A
2007/0185544	12/2006	Dawant et al.	N/A	N/A
2007/0191887	12/2006	Schuler et al.	N/A	N/A
2007/0191912	12/2006	Ficher et al.	N/A	N/A
2007/0197891	12/2006	Shachar et al.	N/A	N/A
2007/0203450	12/2006	Berry	N/A	N/A
2007/0203532	12/2006	Tass et al.	N/A	N/A
2007/0203537	12/2006	Goetz et al.	N/A	N/A
2007/0203538	12/2006	Stone et al.	N/A	N/A
2007/0203539	12/2006	Stone et al.	N/A	N/A
2007/0203540	12/2006	Goetz et al.	N/A	N/A
2007/0203541	12/2006	Goetz et al.	N/A	N/A
2007/0203543	12/2006	Stone et al.	N/A	N/A
2007/0203544	12/2006	Goetz	607/59	A61N 1/37247
2007/0203545	12/2006	Stone et al.	N/A	N/A
2007/0203546	12/2006	Stone et al.	N/A	N/A
2007/0213789	12/2006	Nolan et al.	N/A	N/A
2007/0213790	12/2006	Nolan et al.	N/A	N/A
2007/0244519	12/2006	Keacher et al.	N/A	N/A
2007/0245318	12/2006	Goetz et al.	N/A	N/A
2007/0255321	12/2006	Gerber et al.	N/A	N/A
2007/0255322	12/2006	Gerber et al.	N/A	N/A
2007/0260283	12/2006	Li	N/A	N/A
2007/0265664	12/2006	Gerber et al.	N/A	N/A
2007/0276441	12/2006	Goetz	N/A	N/A
2007/0282189	12/2006	Dan et al.	N/A	N/A
2007/0288064	12/2006	Butson et al.	N/A	N/A
2008/0027514	12/2007	DeMulling et al.	N/A	N/A
2008/0039895	12/2007	Fowler et al.	N/A	N/A
2008/0046037	12/2007	Haubrich et al.	N/A	N/A
2008/0071150	12/2007	Miesel et al.	N/A	N/A
2008/0081982	12/2007	Simon et al.	N/A	N/A
2008/0086451	12/2007	Torres et al.	N/A	N/A
2008/0091248	12/2007	Libbus et al.	N/A	N/A
2008/0103533	12/2007	Patel et al.	N/A	N/A
2008/0114233	12/2007	McIntyre et al.	N/A	N/A
2008/0114579	12/2007	McIntyre et al.	N/A	N/A
2008/0123922	12/2007	Gielen et al.	N/A	N/A

2008/0123923	12/2007	Gielen et al.	N/A	N/A
2008/0133141	12/2007	Frost	N/A	N/A
2008/0141217	12/2007	Goetz et al.	N/A	N/A
2008/0154340	12/2007	Goetz et al.	N/A	N/A
2008/0154341	12/2007	McIntyre et al.	N/A	N/A
2008/0163097	12/2007	Goetz et al.	N/A	N/A
2008/0183256	12/2007	Keacher	N/A	N/A
2008/0188734	12/2007	Suryanarayanan et al.	N/A	N/A
2008/0215101	12/2007	Rezai et al.	N/A	N/A
2008/0215118	12/2007	Goetz et al.	N/A	N/A
2008/0227139	12/2007	Deisseroth et al.	N/A	N/A
2008/0238749	12/2007	Comdorf	N/A	N/A
2008/0242950	12/2007	Jung et al.	N/A	N/A
2008/0261165	12/2007	Steingart et al.	N/A	N/A
2008/0269588	12/2007	Csavoy et al.	N/A	N/A
2008/0300654	12/2007	Lambert et al.	N/A	N/A
2008/0300797	12/2007	Tabibiazar et al.	N/A	N/A
2009/0016491	12/2008	Li	N/A	N/A
2009/0054947	12/2008	Bourn et al.	N/A	N/A
2009/0054950	12/2008	Stephens	N/A	N/A
2009/0082640	12/2008	Kovach et al.	N/A	N/A
2009/0082829	12/2008	Panken et al.	N/A	N/A
2009/0112289	12/2008	Lee et al.	N/A	N/A
2009/0118635	12/2008	Lujan et al.	N/A	N/A
2009/0118786	12/2008	Meadows et al.	N/A	N/A
2009/0149917	12/2008	Whitehurst et al.	N/A	N/A
2009/0163975	12/2008	Gerber et al.	N/A	N/A
2009/0187222	12/2008	Barker	N/A	N/A
2009/0196471	12/2008	Goetz et al.	N/A	N/A
2009/0196472	12/2008	Goetz et al.	N/A	N/A
2009/0198306	12/2008	Goetz et al.	N/A	N/A
2009/0198354	12/2008	Wilson	N/A	N/A
2009/0204192	12/2008	Carlton et al.	N/A	N/A
2009/0208073	12/2008	McIntyre et al.	N/A	N/A
2009/0210208	12/2008	McIntyre et al.	N/A	N/A
2009/0228073	12/2008	Scholten	N/A	N/A
2009/0242399	12/2008	Kamath et al.	N/A	N/A
2009/0270949	12/2008	Kalpin et al.	N/A	N/A
2009/0276008	12/2008	Lee et al.	N/A	N/A
2009/0276021	12/2008	Meadows et al.	N/A	N/A
2009/0281595	12/2008	King et al.	N/A	N/A
2009/0281596	12/2008	King et al.	N/A	N/A
2009/0287271	12/2008	Blum et al.	N/A	N/A
2009/0287272	12/2008	Kokones et al.	N/A	N/A
2009/0287273	12/2008	Carlton et al.	N/A	N/A
2009/0287467	12/2008	Sparks et al.	N/A	N/A
2009/0299164	12/2008	Singhal et al.	N/A	N/A
2009/0299165	12/2008	Singhal et al.	N/A	N/A
2009/0299380	12/2008	Singhal et al.	N/A	N/A
2010/0010387	12/2009	Skelton et al.	N/A	N/A

2010/0010432	12/2009	Skelton	N/A	N/A
2010/0010566	12/2009	Thacker et al.	N/A	N/A
2010/0010646	12/2009	Drew et al.	N/A	N/A
2010/0023103	12/2009	Elborno	N/A	N/A
2010/0023130	12/2009	Henry et al.	N/A	N/A
2010/0030312	12/2009	Shen	N/A	N/A
2010/0049276	12/2009	Blum et al.	N/A	N/A
2010/0049280	12/2009	Goetz	N/A	N/A
2010/0057161	12/2009	Machado et al.	N/A	N/A
2010/0064249	12/2009	Groetken	N/A	N/A
2010/0076535	12/2009	Pianca et al.	N/A	N/A
2010/0113959	12/2009	Pascual-Leon et al.	N/A	N/A
2010/0121409	12/2009	Kothandaraman et al.	N/A	N/A
2010/0135553	12/2009	Joglekar	N/A	N/A
2010/0137944	12/2009	Zhu	N/A	N/A
2010/0152604	12/2009	Kuala et al.	N/A	N/A
2010/0152807	12/2009	Grill et al.	N/A	N/A
2010/0179562	12/2009	Linker et al.	N/A	N/A
2010/0211135	12/2009	Caparso et al.	N/A	N/A
2010/0268298	12/2009	Moffitt	N/A	N/A
2010/0324410	12/2009	Paek et al.	N/A	N/A
2010/0331883	12/2009	Schmitz et al.	N/A	N/A
2011/0004267	12/2010	Meadows	N/A	N/A
2011/0005069	12/2010	Pianca	N/A	N/A
2011/0040351	12/2010	Buston et al.	N/A	N/A
2011/0054559	12/2010	Rosenberg et al.	N/A	N/A
2011/0066407	12/2010	Butson et al.	N/A	N/A
2011/0078900	12/2010	Pianca et al.	N/A	N/A
2011/0093045	12/2010	Moffitt	N/A	N/A
2011/0130803	12/2010	McDonald	N/A	N/A
2011/0130816	12/2010	Howard et al.	N/A	N/A
2011/0130817	12/2010	Chen	N/A	N/A
2011/0130818	12/2010	Chen	N/A	N/A
2011/0137372	12/2010	Makous et al.	N/A	N/A
2011/0160796	12/2010	Lane et al.	N/A	N/A
2011/0172737	12/2010	Davis et al.	N/A	N/A
2011/0184487	12/2010	Alberts et al.	N/A	N/A
2011/0191275	12/2010	Lujan et al.	N/A	N/A
2011/0196253	12/2010	McIntyre et al.	N/A	N/A
2011/0213440	12/2010	Fowler et al.	N/A	N/A
2011/0224665	12/2010	Crosby et al.	N/A	N/A
2011/0224680	12/2010	Barker	N/A	N/A
2011/0238129	12/2010	Moffitt	N/A	N/A
2011/0251583	12/2010	Miyazawa et al.	N/A	N/A
2011/0270348	12/2010	Goetz	N/A	N/A
2011/0306845	12/2010	Osorio	N/A	N/A
2011/0306846	12/2010	Osorio	N/A	N/A
2011/0307032	12/2010	Goetz et al.	N/A	N/A
2011/0313485	12/2010	DeMulling et al.	N/A	N/A
2011/0313500	12/2010	Barker et al.	N/A	N/A

2012/0016378	12/2011	Pianca et al.	N/A	N/A
2012/0027272	12/2011	Akinyemi et al.	N/A	N/A
2012/0046710	12/2011	Digiore et al.	N/A	N/A
2012/0046715	12/2011	Moffitt et al.	N/A	N/A
2012/0071949	12/2011	Pianca et al.	N/A	N/A
2012/0078106	12/2011	Dentinger et al.	N/A	N/A
2012/0089205	12/2011	Boyden et al.	N/A	N/A
2012/0101552	12/2011	Lazarewicz et al.	N/A	N/A
2012/0116476	12/2011	Kothandaraman	N/A	N/A
2012/0165898	12/2011	Moffitt	N/A	N/A
2012/0165901	12/2011	Zhu et al.	N/A	N/A
2012/0165911	12/2011	Pianca	N/A	N/A
2012/0197375	12/2011	Pianca et al.	N/A	N/A
2012/0203316	12/2011	Moffitt et al.	N/A	N/A
2012/0203320	12/2011	Digiore et al.	N/A	N/A
2012/0203321	12/2011	Moffitt et al.	N/A	N/A
2012/0207378	12/2011	Gupta et al.	N/A	N/A
2012/0226138	12/2011	DeSalles et al.	N/A	N/A
2012/0229468	12/2011	Lee et al.	N/A	N/A
2012/0239109	12/2011	Lee	N/A	N/A
2012/0239115	12/2011	Lee	N/A	N/A
2012/0265103	12/2011	Policker et al.	N/A	N/A
2012/0265262	12/2011	Osorio	N/A	N/A
2012/0265268	12/2011	Blum et al.	N/A	N/A
2012/0271189	12/2011	Nelson et al.	N/A	N/A
2012/0277833	12/2011	Gerber et al.	N/A	N/A
2012/0302912	12/2011	Moffitt et al.	N/A	N/A
2012/0303087	12/2011	Moffitt et al.	N/A	N/A
2012/0303098	12/2011	Perryman	N/A	N/A
2012/0314919	12/2011	Sparks et al.	N/A	N/A
2012/0314924	12/2011	Carlton et al.	N/A	N/A
2012/0316615	12/2011	Digiore et al.	N/A	N/A
2012/0316619	12/2011	Goetz et al.	N/A	N/A
2012/0330374	12/2011	Blum et al.	N/A	N/A
2012/0330622	12/2011	Butson et al.	N/A	N/A
2013/0035740	12/2012	Sharma et al.	N/A	N/A
2013/0039550	12/2012	Blum et al.	N/A	N/A
2013/0053926	12/2012	Hincapie Ordonez et al.	N/A	N/A
2013/0060305	12/2012	Bokil	N/A	N/A
2013/0105071	12/2012	Digiore et al.	N/A	N/A
2013/0116744	12/2012	Blum et al.	N/A	N/A
2013/0116748	12/2012	Bokil et al.	N/A	N/A
2013/0116749	12/2012	Carlton et al.	N/A	N/A
2013/0116929	12/2012	Carlton et al.	N/A	N/A
2013/0150922	12/2012	Butson et al.	N/A	N/A
2013/0197424	12/2012	Bedenbaugh	N/A	N/A
2013/0197602	12/2012	Pianca et al.	N/A	N/A
2013/0226261	12/2012	Sparks et al.	N/A	N/A
2013/0245722	12/2012	Ternes et al.	N/A	N/A

2013/0261684	12/2012	Howard	N/A	N/A
2013/0289380	12/2012	Molnar et al.	N/A	N/A
2013/0289660	12/2012	Molnar et al.	N/A	N/A
2013/0317572	12/2012	Zhu et al.	N/A	N/A
2013/0317573	12/2012	Zhu et al.	N/A	N/A
2013/0317587	12/2012	Barker	N/A	N/A
2013/0325091	12/2012	Pianca et al.	N/A	N/A
2014/0039586	12/2013	Barker et al.	N/A	N/A
2014/0039587	12/2013	Romero	N/A	N/A
2014/0063017	12/2013	Kaula et al.	N/A	N/A
2014/0066999	12/2013	Carcieri et al.	N/A	N/A
2014/0067018	12/2013	Carcieri et al.	N/A	N/A
2014/0067022	12/2013	Carcieri et al.	N/A	N/A
2014/0074180	12/2013	Heldman et al.	N/A	N/A
2014/0081366	12/2013	Bentley et al.	N/A	N/A
2014/0107731	12/2013	Stone et al.	N/A	N/A
2014/0122379	12/2013	Moffitt et al.	N/A	N/A
2014/0200633	12/2013	Moffitt	N/A	N/A
2014/0243926	12/2013	Carcieri	N/A	N/A
2014/0276707	12/2013	Jaax	N/A	N/A
2014/0276927	12/2013	Barker	N/A	N/A
2014/0277282	12/2013	Jaax	N/A	N/A
2014/0277284	12/2013	Chen et al.	N/A	N/A
2014/0296953	12/2013	Pianca et al.	N/A	N/A
2014/0343647	12/2013	Romero et al.	N/A	N/A
2014/0353001	12/2013	Romero et al.	N/A	N/A
2014/0358207	12/2013	Romero	N/A	N/A
2014/0358208	12/2013	Howard et al.	N/A	N/A
2014/0358209	12/2013	Romero et al.	N/A	N/A
2014/0358210	12/2013	Howard et al.	N/A	N/A
2015/0018915	12/2014	Leven	N/A	N/A
2015/0021817	12/2014	Romero et al.	N/A	N/A
2015/0045864	12/2014	Howard	N/A	N/A
2015/0051681	12/2014	Hershey	N/A	N/A
2015/0066111	12/2014	Blum et al.	N/A	N/A
2015/0066120	12/2014	Govea	N/A	N/A
2015/0073431	12/2014	Barker	N/A	N/A
2015/0073432	12/2014	Barker	N/A	N/A
2015/0134031	12/2014	Moffitt et al.	N/A	N/A
2015/0151113	12/2014	Govea et al.	N/A	N/A
2015/0246231	12/2014	Martens et al.	N/A	N/A
2015/0360039	12/2014	Lempka et al.	N/A	N/A
2016/0001087	12/2015	Moffitt	N/A	N/A
2016/0008632	12/2015	Wetmore et al.	N/A	N/A
2016/0022995	12/2015	Kothandaraman et al.	N/A	N/A
2016/0023008	12/2015	Kothandaraman	N/A	N/A
2016/0030749	12/2015	Carcieri et al.	N/A	N/A
2016/0030750	12/2015	Bokil et al.	N/A	N/A
2016/0045748	12/2015	Astrom et al.	N/A	N/A
2016/0082252	12/2015	Hershey et al.	N/A	N/A

2016/0096025	12/2015	Moffitt et al.	N/A	N/A
2016/0136429	12/2015	Massoumi et al.	N/A	N/A
2016/0136443	12/2015	Kothandaraman et al.	N/A	N/A
2016/0144186	12/2015	Kaemmerer et al.	N/A	N/A
2016/0175594	12/2015	Min et al.	N/A	N/A
2016/0206380	12/2015	Sparks et al.	N/A	N/A
2016/0228692	12/2015	Steinke et al.	N/A	N/A
2016/0256691	12/2015	Cecchi et al.	N/A	N/A
2016/0256693	12/2015	Parramon	N/A	N/A
2016/0317800	12/2015	Barker	N/A	N/A
2016/0346557	12/2015	Bokil	N/A	N/A
2016/0375248	12/2015	Carcieri et al.	N/A	N/A
2016/0375258	12/2015	Steinke	N/A	N/A
2017/0100593	12/2016	Zottola	N/A	N/A
2017/0100601	12/2016	Xiao et al.	N/A	N/A
2017/0106197	12/2016	Wechter et al.	N/A	N/A
2017/0197086	12/2016	Howard et al.	N/A	N/A
2017/0225007	12/2016	Orinski	N/A	N/A
2017/0252570	12/2016	Serrano Carmona et al.	N/A	N/A
2017/0259078	12/2016	Howard	N/A	N/A
2017/0304610	12/2016	Huibregtse et al.	N/A	N/A
2017/0304633	12/2016	Zhang	N/A	N/A
2017/0372039	12/2016	Mustakos et al.	N/A	N/A
2018/0064930	12/2017	Zhang et al.	N/A	N/A
2018/0078776	12/2017	Mustakos et al.	N/A	N/A
2018/0104482	12/2017	Bokil	N/A	N/A
2018/0104500	12/2017	Blum	N/A	A61N 1/36125
2018/0110971	12/2017	Serrano Carmona	N/A	N/A
2018/0133481	12/2017	Von Zitzewitz et al.	N/A	N/A
2018/0185650	12/2017	Shah	N/A	N/A
2018/0193655	12/2017	Zhang et al.	N/A	N/A
2018/0214700	12/2017	Vansickle et al.	N/A	N/A
2018/0264278	12/2017	Laghi	N/A	N/A
2018/0272142	12/2017	Zhang et al.	N/A	N/A
2018/0280698	12/2017	Steinke et al.	N/A	N/A
2018/0296828	12/2017	Bradley et al.	N/A	N/A
2018/0333173	12/2017	Wang	N/A	N/A
2018/0369589	12/2017	Schouenborg	N/A	A61N 1/36132
2018/0369606	12/2017	Zhang et al.	N/A	N/A
2018/0369607	12/2017	Zhang et al.	N/A	N/A
2019/0015660	12/2018	Zhang et al.	N/A	N/A
2019/0105503	12/2018	Leven	N/A	N/A
2019/0134403	12/2018	Steinke	N/A	A61N 1/37247
2019/0184167	12/2018	Vansickle	N/A	A61N 1/36185
2019/0184171	12/2018	Mustakos et al.	N/A	N/A
2019/0209834	12/2018	Zhang et al.	N/A	N/A

2019/0209849	12/2018	Hershey et al.	N/A	N/A
2019/0262609	12/2018	Brill et al.	N/A	N/A
2019/0275331	12/2018	Zhu	N/A	N/A
2019/0329024	12/2018	Kothandaraman	N/A	A61N 1/36171
2019/0329047	12/2018	Moffitt et al.	N/A	N/A
2019/0329049	12/2018	Carcieri et al.	N/A	N/A
2020/0094047	12/2019	Govea et al.	N/A	N/A
2020/0139127	12/2019	Zhang et al.	N/A	N/A
2020/0155854	12/2019	Leven et al.	N/A	N/A
2020/0155859	12/2019	Blum et al.	N/A	N/A
2020/0171298	12/2019	Goetz et al.	N/A	N/A
2020/0171310	12/2019	Walter et al.	N/A	N/A
2020/0179600	12/2019	Zanos et al.	N/A	N/A
2020/0179688	12/2019	Su	N/A	A61N 1/025
2020/0215330	12/2019	Huertas Fernandez et al.	N/A	N/A
2020/0222704	12/2019	Moffitt et al.	N/A	N/A
2020/0269053	12/2019	Park	N/A	N/A
2020/0353254	12/2019	O Laighin et al.	N/A	N/A
2020/0376262	12/2019	Clark et al.	N/A	N/A
2020/0376263	12/2019	Zhu	N/A	N/A
2020/0398057	12/2019	Esteller et al.	N/A	N/A
2021/0008388	12/2020	Vansickle et al.	N/A	N/A
2021/0008389	12/2020	Featherstone et al.	N/A	N/A
2021/0016111	12/2020	Vansickle et al.	N/A	N/A
2021/0023374	12/2020	Block et al.	N/A	N/A
2021/0052893	12/2020	Suri et al.	N/A	N/A
2021/0113844	12/2020	Zhang et al.	N/A	N/A
2021/0128920	12/2020	Grill et al.	N/A	N/A
2021/0196956	12/2020	Juárez Paz	N/A	N/A
2021/0196964	12/2020	Schnell	N/A	A61B 5/372
2021/0205613	12/2020	Bradley et al.	N/A	N/A
2021/0268268	12/2020	Horn et al.	N/A	N/A
2021/0275820	12/2020	Grill, Jr. et al.	N/A	N/A
2021/0387002	12/2020	Bourget et al.	N/A	N/A
2022/0007980	12/2021	Single	N/A	N/A
2022/0008729	12/2021	Zhu	N/A	N/A
2022/0040485	12/2021	Li et al.	N/A	N/A
2022/0062640	12/2021	Raike	N/A	A61N 1/36067
2022/0072329	12/2021	Howard	N/A	N/A
2022/0111213	12/2021	Cassar	N/A	A61N 1/36132
2022/0126100	12/2021	Jackson et al.	N/A	N/A
2022/0257950	12/2021	Moore et al.	N/A	N/A
2022/0266026	12/2021	Case et al.	N/A	N/A
2022/0296892	12/2021	Esteller et al.	N/A	N/A
2022/0296893	12/2021	Steinke et al.	N/A	N/A
2022/0339448	12/2021	Jayakumar et al.	N/A	N/A

2022/0347479	12/2021	Esteller et al.	N/A	N/A
2022/0355114	12/2021	Moore et al.	N/A	N/A
2022/0355115	12/2021	Moore et al.	N/A	N/A
2022/0370808	12/2021	Esteller	N/A	N/A
2022/0387785	12/2021	Huynh et al.	N/A	N/A
2022/0395690	12/2021	Haddock et al.	N/A	N/A
2023/0048571	12/2022	Poltorak	N/A	N/A
2023/0050186	12/2022	Rotenstreich	N/A	A61B 5/7267
2023/0064552	12/2022	Moffitt	N/A	N/A
2023/0141183	12/2022	Moore et al.	N/A	N/A
2023/0181089	12/2022	Zhang et al.	N/A	N/A
2023/0181090	12/2022	Juarez Paz	N/A	N/A
2023/0181906	12/2022	Moore et al.	N/A	N/A
2023/0248977	12/2022	Esteller et al.	N/A	N/A
2023/0264025	12/2022	Malekmohammadi et al.	N/A	N/A
2023/0271015	12/2022	Malekmohammadi et al.	N/A	N/A
2023/0277854	12/2022	Gaviao Kilmar	N/A	N/A
2024/0065620	12/2023	Moore et al.	N/A	N/A
2024/0157151	12/2023	Juarez Paz	N/A	N/A
2024/0198110	12/2023	Moore	N/A	N/A
2024/0316346	12/2023	Shah et al.	N/A	N/A
2024/0359015	12/2023	Steinke et al.	N/A	N/A
2025/0010079	12/2024	Bokil	N/A	N/A
2025/0050107	12/2024	Moore et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
813889	12/1996	EP	N/A
1048320	12/1999	EP	N/A
1166819	12/2001	EP	N/A
1372780	12/2003	EP	N/A
1559369	12/2004	EP	N/A
97/39797	12/1996	WO	N/A
98/48880	12/1997	WO	N/A
01/90876	12/2000	WO	N/A
02/26314	12/2001	WO	N/A
02/28473	12/2001	WO	N/A
02/065896	12/2001	WO	N/A
02/072192	12/2001	WO	N/A
03/086185	12/2002	WO	N/A
2004/019799	12/2003	WO	N/A
2004041080	12/2004	WO	N/A
2006017053	12/2005	WO	N/A
2006113305	12/2005	WO	N/A
20071097859	12/2006	WO	N/A
20071097861	12/2006	WO	N/A

2007/100427	12/2006	WO	N/A
2007/100428	12/2006	WO	N/A
2007/112061	12/2006	WO	N/A
2009097224	12/2008	WO	N/A
2010/109448	12/2009	WO	N/A
2010/120823	12/2009	WO	N/A
2011025865	12/2010	WO	N/A
2011/139779	12/2010	WO	N/A
2011/159688	12/2010	WO	N/A
2012088482	12/2011	WO	N/A
2016/025913	12/2015	WO	N/A
2016081099	12/2015	WO	N/A
2016112398	12/2015	WO	N/A

OTHER PUBLICATIONS

Zhang, Y., et al., "Atlas-guided tract reconstruction for automated and comprehensive examination of the white matter anatomy," Neuroimage 52(4) (2010), pp. 1289-1301. cited by applicant

"BioPSE" The Biomedical Problem Solving Environment",
<http://www.sci.utah.edu/cibc/software/index.html>, MCRR Center for Integrative Biomedical Computing,(2004). cited by applicant

Andrews, R. J., "Neuroprotection trek-the next generation: neuromodulation I. Techniques—deep brain stimulation, vagus nerve stimulation, and transcranial magnetic stimulation.", Ann NY Acad Sci. 993. (May 2003),1-13. cited by applicant

Carnevale, N.T. et al., "The Neuron Book," Cambridge, UK: Cambridge University Press (2006), 480 pages. cited by applicant

Chaturvedi: "Development of Accurate Computational Models for Patient-Specific Deep Brain Stimulation," Electronic Thesis or Dissertation, Jan. 2012, 162 pages. cited by applicant

Chaturvedi, A. et al.: "Patient-specific models of deep brain stimulation: Influence of field model complexity on neural activation predictions." Brain Stimulation, Elsevier, Amsterdam, NL, vol. 3, No. Apr. 2, 2010, pp. 65-77. cited by applicant

Frankemolle, et al., "Reversing cognitive-motor impairments in Parkinson's disease patients using a computational modeling approach to deep brain stimulation programming," Brian 133 (2010), pp. 746-761. cited by applicant

McIntyre, C.C., et al., "Modeling the excitability of mammalian nerve fibers: influence of afterpotentials on the recovery cycle," J Neurophysiol, 87(2) (Feb. 2002), pp. 995-1006. cited by applicant

Peterson, et al., "Predicting myelinated axon activation using spatial characteristics of the extracellular field," Journal of Neural Engineering, 8 (2011), 12 pages. cited by applicant

Warman, et al., "Modeling the Effects of Electric Fields on nerver Fibers; Dermination of Excitation Thresholds," IEEE Transactions on Biomedical Engineering, vol. 39, No. 12 (Dec. 1992), pp. 1244-1254. cited by applicant

Wesselink, et al., "Analysis of Current Density and Related Parameters in Spinal Cord Stimulation," IEEE Transactions on Rehabilitation Engineering, vol. 6, No. Jun. 2, 1998, pp. 200-207. cited by applicant

Andrews, R. J., "Neuroprotection trek-the next generation: neuromodulation II. Applications—epilepsy, nerve regeneration, neurotrophins.", Ann NY Acad Sci. 993 (May 2003), 14-24. cited by applicant

Astrom, M. , et al., "The effect of cystic cavities on deep brain stimulation in the basal ganglia: a simulation-based study", J Neural Eng., 3(2), (Jun. 2006).132-8. cited by applicant

Bazin et al., "Free Software Tools for Atlas-based Volumetric Neuroimage Analysis", Proc. SPIE

5747, Medical Imaging 2005: Image Processing, 1824 May 5, 2005. cited by applicant

Back, C. , et al., "Postoperative Monitoring of the Electrical Properties of Tissue and Electrodes in Deep Brain Stimulation", Neuromodulation, 6(4), (Oct. 2003),248-253. cited by applicant

Baker, K. B., et al., "Evaluation of specific absorption rate as a dosimeter of MRI-related implant heating", J Magn Reson Imaging., 20(2), (Aug. 2004),315-20. cited by applicant

Brown, J. "Motor Cortex Stimulation," Neurosurgical Focus (Sep. 15, 2001) 11(3):E5. cited by applicant

Budai et al., "Endogenous Opioid Peptides Acting at m-Opioid Receptors in the Dorsal Horn Contribute to Midbrain Modulation of Spinal Nociceptive Neurons, " Journal of Neurophysiology (1998) 79(2): 677-687. cited by applicant

Cesselin, F. "Opioid and anti-opioid peptides," Fundamental and Clinical Pharmacology (1995) 9(5): 409-33 (Abstract only). cited by applicant

Rezai et al., "Deep Brain Stimulation for Chronic Pain" Surgical Management of Pain, Chapter 44 pp. 565-576 (2002). cited by applicant

Xu, MD., Shi-Ang, article entitled "Comparison of Half-Band and Full-Band Electrodes for Intracochlear Electrical Stimulation", Annals of Otology, Rhinology & Laryngology (Annals of Head & Neck Medicine & Surgery), vol. 102 (5) pp. 363-367 May 1993. cited by applicant

Bedard, C. , et al., "Modeling extracellular field potentials and the frequency-filtering properties of extracellular space", Biophys J . . . 86(3). (Mar. 2004),1829-42. cited by applicant

Benabid, A. L., et al., "Future prospects of brain stimulation", Neurol Res.;22(3), (Apr. 2000),237-46. cited by applicant

Brummer, S. B., et al., "Electrical Stimulation with Pt Electrodes: II—Estimation of Maximum Surface Redox (Theoretical Non-Gassing) Limits", IEEE Transactions on Biomedical Engineering, vol. BME-24, Issue 5, (Sep. 1977),440-443. cited by applicant

Butson, Christopher R., et al., "Deep Brain Stimulation of the Subthalamic Nucleus: Model-Based Analysis of the Effects of Electrode Capacitance on the vol. of Activation", Proceedings of the 2nd International IEEE EMBS, (Mar. 16-19, 2005),196-197. cited by applicant

Mcintyre, Cameron C., et al., "Cellular effects of deep brain stimulation: model-based analysis of activation and Inhibition," J Neurophysiol, 91(4) (Apr. 2004), pp. 1457-1469. cited by applicant

Chaturvedi, A., et al., "Subthalamic Nucleus Deep Brain Stimulation: Accurate Axonal Threshold Prediction with Diffusion Tensor Based Electric Field Models", Engineering in Medicine and Biology Society, 2006. EMBS' 06 28th Annual International Conference of the IEEE, IEEE, Piscataway, NJ USA, Aug. 30, 2006. cited by applicant

Butson, Christopher et al., "Predicting the Effects of Deep Brain Stimulation with Diffusion Tensor Based Electric Field Models" Jan. 1, 2001, Medical Image Computing and Computer-Assisted Intervention-Mic CAI 2006 Lecture Notes in Computer Science; LNCS, Springer, Berlin, DE. cited by applicant

Butson, C. R., et al., "Deep brainstimulation interactive visualization system", Society for Neuroscience vol. 898.7 (2005). cited by applicant

Hodaie, M., et al., "Chronic anterior thalamus stimulation for intractable epilepsy," Epilepsia, 43(6) (Jun. 2002), pp. 603-608. cited by applicant

Hoekema, R., et al., "Multigrid solution of the potential field in modeling electrical nerve stimulation," Comput Biomed Res., 31(5) (Oct. 1998), pp. 348-362. cited by applicant

Holsheimer, J., et al., "Identification of the target neuronal elements in electrical deep brain stimulation," Eur J Neurosci., 12(12) (Dec. 2000), pp. 4573-4577. cited by applicant

Jezernik, S., et al., "Neural network classification of nerve activity recorded in a mixed nerve," Neurol Res., 23(5) (Jul. 2001), pp. 429-434. cited by applicant

Jones, DK., et al., "Optimal strategies for measuring diffusion in anisotropic systems by magnetic resonance imaging," Magn. Reson. Med., 42(3) (Sep. 1999), pp. 515-525. cited by applicant

Krack, P., et al., "Postoperative management of subthalamic nucleus stimulation for Parkinson's

disease,” *Mov. Disord.*, vol. 17(suppl 3) (2002), pp. 188-197. cited by applicant

Le Bihan, D., et al., “Diffusion tensor imaging: concepts and applications,” *J Magn Reson Imaging*, 13(4) (Apr. 2001), pp. 534-546. cited by applicant

Lee, D. C., et al., “Extracellular electrical stimulation of central neurons: quantitative studies,” In: *Handbook of neuroprosthetic methods*, WE Finn and PG Lopresti (eds) CRC Press (2003), pp. 95-125. cited by applicant

Levy, AL., et al., “An Internet-connected, patient-specific, deformable brain atlas integrated into a surgical navigation system,” *J Digit Imaging*, 10(3 Suppl 1) (Aug. 1997), pp. 231-237. cited by applicant

Liu, Haiying, et al., “Intra-operative MR-guided DBS implantation for treating PD and ET,” *Proceedings of SPIE* vol. 4319, Department of Radiology & Neurosurgery, University of Minnesota, Minneapolis, MN 55455 (2001), pp. 272-276. cited by applicant

Mcintyre, C. C., et al., “Extracellular stimulation of central neurons: influence of stimulus waveform and frequency on neuronal output,” *J. Neurophysiol.*, 88(4), (Oct. 2002), pp. 1592-1604. cited by applicant

Mcintyre, C. C., et al., “Microstimulation of spinal motoneurons: a model study,” *Proceedings of the 19th Annual International Conference of the IEEE Engineering in Medicine and Biology society*, vol. 5, (1997), pp. 2032-2034. cited by applicant

Mcintyre, Cameron C., et al., “Model-based Analysis of deep brain stimulation of the thalamus,” *Proceedings of the Second joint EMBS/BM ES Conference*, vol. 3, Annual Fall Meeting of the Biomedical Engineering Society (Cal. No. 02CH37392) IEEE Piscataway, NJ (2002), pp. 2047-2048. cited by applicant

Mcintyre, C. C., et al., “Model-based design of stimulus trains for selective microstimulation of targeted neuronal populations,” *Proceedings of the 23rd Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, vol. 1 (2001), pp. 806-809. cited by applicant

Mcintyre, C. C., et al., Model-based design of stimulus waveforms for selective microstimulation in the central nervous system,, *Proceedings of the First Joint [Engineering in Medicine and Biology, 1999. 21st Annual Conf. and the 1999 Annual Fall Meeting of the Biomedical Engineering Soc.] BM ES/EMBS Conference*, vol. 1 (1999), p. 384. cited by applicant

Mcintyre, Cameron C., et al., “Selective microstimulation of central nervous system neurons,” *Annals of biomedical engineering*, 28(3) (Mar. 2000), pp. 219-233. cited by applicant

Mcintyre, C. C., et al., “Sensitivity analysis of a model of mammalian neural membrane,” *Biol Cybern.*, 79(1) (Jul. 1998), pp. 29-37. cited by applicant

Mcintyre, Cameron C., et al., “Uncovering the mechanism(s) of action of deep brain stimulation: activation, inhibition, or both,” *Clin Neurophysiol*, 115(6) (Jun. 2004), pp. 1239-1248. cited by applicant

Mcintyre, Cameron C., et al., “Uncovering the mechanisms of deep brain stimulation for Parkinson's disease through functional imaging, neural recording, and neural modeling,” *Crit Rev Biomed Eng.*, 30(4-6) (2002), pp. 249-281. cited by applicant

Mouine et al. “Multi-Strategy and Multi-Algorithm Cochlear Prostheses”, *Biomed. Sci. Instrument*, 2000; 36:233-238. cited by applicant

Struijk, J J., et al., “Recruitment of dorsal column fibers in spinal cord stimulation: influence of collateral branching,” *IEEE Transactions on Biomedical Engineering*, 39(9) (Sep. 1992), pp. 903-912. cited by applicant

Nakana, S., et al., “Fiber tract-based atlas of human white matter anatomy,” *Radiology*, 230(1) (Jan. 2004), pp. 77-87. cited by applicant

Voghell et al., “Programmable Current Source Dedicated to Implantable Microstimulators” *ICM '98 Proceedings of the Tenth International Conference*, pp. 67-70. cited by applicant

Butson, Christopher R., et al., “Patient-specific analysis of the volume of tissue activated during deep brain stimulation”, *NeuroImage*. vol. 34. (2007),661-670. cited by applicant

Alo, K. M., et al., "New trends in neuromodulation for the management of neuropathic pain," *Neurosurgery*, 50(4), (Apr. 2002), pp. 690-703, discussion pp. 703-704. cited by applicant

Ashby, P., et al., "Neurophysiological effects of stimulation through electrodes in the human subthalamic nucleus," *Brain*, 122 (Pt 10), (Oct. 1999), pp. 1919-1931. cited by applicant

Baker, K. B., et al., "Subthalamic nucleus deep brain stimulus evoked potentials: Physiological and therapeutic implications," *Movement Disorders*, 17(5), (Sep./Oct. 2002), pp. 969-983. cited by applicant

Bammer, R, et al., "Diffusion tensor imaging using single-shot SENSE-EPI", *Magn Reson Med.*, 48(1), (Jul. 2002), pp. 128-136. cited by applicant

Basser, P J., et al., "MR diffusion tensor spectroscopy and imaging," *Biophys J.*, 66(1), (Jan. 1994), pp. 259-267. cited by applicant

Basser, P J., et al., "New currents in electrical stimulation of excitable tissues," *Annu Rev Biomed Eng.*, 2, (2000), pp. 377-397. cited by applicant

Benabid, AL., et al., "Chronic electrical stimulation of the ventralis intermedius nucleus of the thalamus as a treatment of movement disorders," *J. Neurosurg.*, 84(2), (Feb. 1996), pp. 203-214. cited by applicant

Benabid, AL., et al., "Combined (thalamotomy and stimulation) stereotactic surgery of the VIM thalamic nucleus for bilateral Parkinson disease," *Appl Neurophysiol*, vol. 50, (1987), pp. 344-346. cited by applicant

Benabid, A L., et al., "Long-term suppression of tremor by chronic stimulation of the ventral intermediate thalamic nucleus," *Lancet*, 337 (8738), (Feb. 16, 1991), pp. 403-406. cited by applicant

Butson, C. R., et al., "Predicting the effects of deep brain stimulation with diffusion tensor based electric field models," *Medical Image Computing and Computer-Assisted Intervention—Mic Cai 2006, Lecture Notes in Computer Science (LNCS)*, vol. 4191, pp. 429-437, LNCS, Springer, Berlin, DE. cited by applicant

Christensen, Gary E., et al., "Volumetric transformation of brain anatomy," *IEEE Transactions on Medical Imaging*, 16 (6), (Dec. 1997), pp. 864-877. cited by applicant

Cooper, S , et al., "Differential effects of thalamic stimulation parameters on tremor and paresthesias in essential tremor," *Movement Disorders*, 17(Suppl. 5), (2002), p. S193. cited by applicant

Coubes, P, et al., "Treatment of DYT1-generalised dystonia by stimulation of the internal globus pallidus," *Lancet*, 355 (9222), (Jun. 24, 2000), pp. 2220-2221. cited by applicant

Dasilva, A.F. M., et al., "A Primer Diffusion Tensor Imaging of Anatomical Substructures," *Neurosurg. Focus*; 15(1) (Jul. 2003), pp. 1-4. cited by applicant

Dawant, B. M., et al., "Computerized atlas-guided positioning of deep brain stimulators: a feasibility study," *Biomedical Image registration, Second International Workshop, WBIR 2003, Revised Papers (Lecture notes in Comput. Sci. vol. (2717), Springer-Verlag Berlin, Germany(2003)*, pp. 142-150. cited by applicant

Finnis, K. W., et al., "3-D functional atlas of subcortical structures for image guided stereotactic neurosurgery," *Neuroimage*, vol. 9, No. 6, Iss. 2 (1999), p. S206. cited by applicant

Finnis, K. W., et al., "3D Functional Database of Subcortical Structures for Surgical Guidance in Image Guided Stereotactic Neurosurgery," *Medical Image Computing and Computer-Assisted Intervention-MICCAI'99, Second International Conference, Cambridge, UK, Sep. 19-22, 1999, Proceedings (1999)*, pp. 758-767. cited by applicant

Finnis, K. W., et al., "A 3-Dimensional Database of Deep Brain Functional Anatomy, and Its Application to Image-Guided Neurosurgery," *Proceedings of the Third International Conference on Medical Image Computing and Computer-Assisted Intervention, Lecture Notes In Computer Science*; vol. 1935 (2000), pp. 1-8. cited by applicant

Finnis, K. W., et al., "A functional database for guidance of surgical and therapeutic procedures in

the deep brain,” Proceedings of the 22nd Annual International Conference of the IEEE Engineering in Medicine and Biology Society, vol. 3 (2000), pp. 1787-1789. cited by applicant

Finnis, K. W., et al., “Application of a Population Based Electrophysiological Database to the Planning and Guidance of Deep Brain Stereotactic Neurosurgery,” Proceedings of the 5th International Conference on Medical Image Computing and Computer-Assisted Intervention—Part 11, Lecture Notes In Computer Science; vol. 2489 (2002), pp. 69-76. cited by applicant

Finnis, K. W., et al., “Subcortical physiology deformed into a patient-specific brain atlas for image-guided stereotaxy,” Proceedings of SPIE—vol. 4681 Medical Imaging 2002: Visualization, Image-Guided Procedures, and Display (May 2002), pp. 184-195. cited by applicant

Finnis, Krik W., et al., “Three-Dimensional Database of Subcortical Electrophysiology for Image-Guided Stereotatic Functional Neurosurgery,” IEEE Transactions on Medical Imaging, 22(1) (Jan. 2003), pp. 93-104. cited by applicant

Gabriels, L , et al., “Deep brain stimulation for treatment-refractory obsessive-compulsive disorder: psychopathological and neuropsychological outcome in three cases,” Acta Psychiatr Scand., 107(4) (2003), pp. 275-282. cited by applicant

Gabriels, LA., et al., “Long-term electrical capsular stimulation in patients with obsessive-compulsive disorder,” Neurosurgery, 52(6) (Jun. 2003), pp. 1263-1276. cited by applicant

Goodall, E. V., et al., “Modeling study of activation and propagation delays during stimulation of peripheral nerve fibers with a tripolar cuff electrode,” IEEE Transactions on Rehabilitation Engineering, [see also IEEE Trans. on Neural Systems and Rehabilitation], 3(3) (Sep. 1995), pp. 272-282. cited by applicant

Goodall, E. V., et al., “Position-selective activation of peripheral nerve fibers with a cuff electrode,” IEEE Transactions on Biomedical Engineering, 43(8) (Aug. 1996), pp. 851-856. cited by applicant

Goodall, E. V., “Simulation of activation and propagation delay during tripolar neural stimulation,” Proceedings of the 15th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (1993), pp. 1203-1204. cited by applicant

Grill, WM., “Modeling the effects of electric fields on nerve fibers: influence of tissue electrical properties,” IEEE Transactions on Biomedical Engineering, 46(8) (1999), pp. 918-928. cited by applicant

Grill, W. M., et al., “Neural and connective tissue response to long-term implantation of multiple contact nerve cuff electrodes,” J Biomed Mater Res., 50(2) (May 2000), pp. 215-226. cited by applicant

Grill, W. M., “Neural modeling in neuromuscular and rehabilitation research,” Proceedings of the 23rd Annual International Conference of the IEEE Engineering in Medicine and Biology Society, vol. 4 (2001), pp. 4065-4068. cited by applicant

Grill, W. M., et al., “Non-invasive measurement of the input-output properties of peripheral nerve stimulating electrodes,” Journal of Neuroscience Methods, 65(1) (Mar. 1996), pp. 43-50. cited by applicant

Grill, W. M., et al., “Quantification of recruitment properties of multiple contact cuff electrodes,” IEEE Transactions on Rehabilitation Engineering, [see also IEEE Trans. on Neural Systems and Rehabilitation], 4(2) (Jun. 1996), pp. 49-62. cited by applicant

Grill, W. M., “Spatially selective activation of peripheral nerve for neuroprosthetic applications,” Ph.D. Case Western Reserve University, (1995), pp. 245 pages. cited by applicant

Grill, W. M., “Stability of the input-output properties of chronically implanted multiple contact nerve cuff stimulating electrodes,” IEEE Transactions on Rehabilitation Engineering [see also IEEE Trans. on Neural Systems and Rehabilitation] (1998), pp. 364-373. cited by applicant

Grill, W. M., “Stimulus waveforms for selective neural stimulation,” IEEE Engineering in Medicine and Biology Magazine, 14(4) (Jul.-Aug. 1995), pp. 375-385. cited by applicant

Grill, W. M., et al., “Temporal stability of nerve cuff electrode recruitment properties,” IEEE 17th Annual Conference Engineering in Medicine and Biology Society, vol. 2 (1995), pp. 1089-1090.

cited by applicant

Gross, RE., et al., "Advances in neurostimulation for movement disorders," *Neurol Res.*, 22(3) (Apr. 2000), pp. 247-258. cited by applicant

Guridi et al., "The subthalamic nucleus, hemiballismus and Parkinson's disease: reappraisal of a neurological dogma," *Brain*, vol. 124, 2001, pp. 5-19. cited by applicant

Haberler, C, et al., "No tissue damage by chronic deep brain stimulation in Parkinson's disease," *Ann Neurol.*, 48(3) (Sep. 2000), pp. 372-376. cited by applicant

Hamel, W, et al., "Deep brain stimulation of the subthalamic nucleus in Parkinson's disease: evaluation of active electrode contacts," *J Neurol Neurosurg Psychiatry*, 74(8) (Aug. 2003), pp. 1036-1046. cited by applicant

Hanekom, "Modelling encapsulation tissue around cochlear implant electrodes," *Med. Biol. Eng. Comput.* vol. 43 (2005), pp. 47-55. cited by applicant

Haueisen, J , et al., "The influence of brain tissue anisotropy on human EEG and MEG," *Neuroimage*, 15(1) (Jan. 2002), pp. 159-166. cited by applicant

D'Haese et al. *Medical Image Computing and Computer-Assisted Intervention—MICCAI 2005 Lecture Notes in Computer Science*, 2005, vol. 3750, 2005, 427-434. cited by applicant

Rohde et al. *IEEE Transactions on Medical Imaging*, vol. 22 No. 11, 2003 p. 1470-1479. cited by applicant

Dawant et al., *Biomedical Image Registration. Lecture Notes in Computer Science*, 2003, vol. 2717, 2003, 142-150. cited by applicant

Miocinovic et al., "Stereotactic Neurosurgical Planning, Recording, and Visualization for Deep Brain Stimulation in Non-Human Primates", *Journal of Neuroscience Methods*, 162:32-41, Apr. 5, 2007, XP022021469. cited by applicant

Gemmar et al., "Advanced Methods for Target Navigation Using Microelectrode Recordings in Stereotactic Neurosurgery for Deep Brain Stimulation", *21st IEEE International Symposium on Computer-Based Medical Systems*, Jun. 17, 2008, pp. 99-104, XP031284774. cited by applicant

Acar et al., "Safety Anterior Commissure-Posterior Commissure-Based Target Calculation of the Subthalamic Nucleus in Functional Stereotactic Procedures", *Stereotactic Funct. Neurosurg.*, 85:287-291, Aug. 2007. cited by applicant

Andrade-Souza, "Comparison of Three Methods of Targeting the Subthalamic Nucleus for Chronic Stimulation in Parkinson's Disease", *Neurosurgery*, 56:360-368, Apr. 2005. cited by applicant

Anheim et al., "Improvement in Parkinson Disease by Subthalamic Nucleus Stimulation Based on Electrode Placement", *Arch Neural.*, 65:612-616, May 2008. cited by applicant

Butson et al., "Tissue and Electrode Capacitance Reduce Neural Activation vols. During Deep Brain Stimulation", *Clinical Neurophysiology*, 116:2490-2500, Oct. 2005. cited by applicant

Butson et al., "Sources and Effects of Electrode Impedance During Deep Brain Stimulation", *Clinical Neurophysiology*, 117:44 7-454, Dec. 2005. cited by applicant

D'Haese et al., "Computer-Aided Placement of Deep Brain Stimulators: From Planning to Intraoperative Guidance", *IEEE Transaction on Medical Imaging*, 24:1469-1478, Nov. 2005. cited by applicant

Gross et al., "Electrophysiological Mapping for the Implantation of Deep Brain Stimulators for Parkinson's Disease and Tremor", *Movement Disorders*, 21 :S259-S283, Jun. 2006. cited by applicant

Halpern et al., "Brain Shift During Deep Brain Stimulation Surgery for Parkinson's Disease", *Stereotact Funct. Neurosurg.*, 86:37-43, published online Sep. 2007. cited by applicant

Herzog et al., "Most Effective Stimulation Site in Subthalamic Deep Brain Stimulation for Parkinson's Disease", *Movement Disorders*, 19:1050-1099, published on line Mar. 2004. cited by applicant

Jeon et al., *A Feasibility Study of Optical Coherence Tomography for Guiding Deep Brain Probes*, *Journal of Neuroscience Methods*, 154:96-101, Jun. 2006. cited by applicant

Khan et al., "Assessment of Brain Shift Related to Deep Brain Stimulation Surgery", *Stereotact Funct. Neurosurg.*, 36:44-53, published online Sep. 2007. cited by applicant

Koop et al., "Improvement in a Quantitative Measure of Bradykinesia After Microelectrode Recording in Patients with Parkinson's Disease During Deep Brain Stimulation Surgery", *Movement Disorders*, 21 :673-678, published on line Jan. 2006. cited by applicant

Lemaire et al., "Brain Mapping in Stereotactic Surgery: A Brief Overview from the Probabilistic Targeting to the Patient-Based Anatomic Mapping", *NeuroImage*, 37:S109-S115, available online Jun. 2007. cited by applicant

Machado et al., "Deep Brain Stimulation for Parkinson's Disease: Surgical Technique and Perioperative Management", *Movement Disorders*, 21 :S247-S258, Jun. 2006. cited by applicant

Maks et al., "Deep Brain Stimulation Activation Volumes and Their Association with Neurophysiological Mapping and Therapeutic Outcomes", Downloaded from jnnp.bmj.com, pp. 1-21, published online Apr. 2008. cited by applicant

Moran et al., "Real-Time Refinement of Subthalamic Nucleus Targeting Using Bayesian Decision-Making on the Root Mean Square Measure", *Movement Disorders*, 21: 1425-1431, published online Jun. 2006. cited by applicant

Sakamoto et al., "Homogeneous Fluorescence Assays for RNA Diagnosis by Pyrene-Conjugated 2'-O-Methyloligoribonucleotides", *Nucleosides, Nucleotides, and Nuclerics Acids*, 26:1659-1664, on line publication Oct. 2007. cited by applicant

Winkler et al., The First Evaluation of Brain Shift During Functional Neurosurgery by Deformation Field Analysis, *J. Neural. Neurosurg. Psychiatry*, 76:1161-1163, Aug. 2005. cited by applicant

Yelnik et al., "A Three-Dimensional, Histological and Deformable Atlas of the Human Basal J Ganglia. I. Atlas Construction Based on Immunohistochemical and MRI Data", *NeuroImage*, 34:618,-638,Jan. 2007. cited by applicant

Ward, H. E., et al., "Update on deep brain stimulation for neuropsychiatric disorders," *Neurobiol Dis* 38 (3) (2010), pp. 346-353. cited by applicant

Alberts et al. "Bilateral subthalamic stimulation impairs cognitive-motor performance in Parkinson's disease patients." *Brain* (2008), 131, 3348-3360, Abstract. cited by applicant

An, et al., "Prefrontal cortical projections to longitudinal columns in the midbrain periaqueductal gray in macaque monkeys," *J Comp Neural* 401 (4) (1998), pp. 455-479. cited by applicant

Carmichael, S. T., et al., "Connectional networks within the orbital and medial prefrontal cortex of macaque monkeys," *J Comp Neural* 371 (2) (1996), pp. 179-207. cited by applicant

Croxson, et al., "Quantitative investigation of connections of the prefrontal cortex in the human and macaque using probabilistic diffusion tractography," *J Neurosci* 25 (39) (2005), pp. 8854-8866. cited by applicant

Frankemolle, et al., "Reversing cognitive-motor impairments in Parkinson's disease patients using a computational modelling approach to deep brain stimulation programming," *Brain* 133 (2010), pp. 746-761. cited by applicant

Freedman, et al., "Subcortical projections of area 25 (subgenual cortex) of the macaque monkey," *J Comp Neurol* 421 (2) (2000), pp. 172-188. cited by applicant

Giacobbe, et al., "Treatment resistant depression as a failure of brain homeostatic mechanisms: implications for deep brain stimulation," *Exp Neural* 219 (1) (2009), pp. 44-52. cited by applicant

Goodman, et al., "Deep brain stimulation for intractable obsessive compulsive disorder: pilot study using a blinded, staggered-onset design," *Biol Psychiatry* 67 (6) (2010), pp. 535-542. cited by applicant

Greenberg, et al., "Deep brain stimulation of the ventral internal capsule/ventral striatum for obsessive-compulsive disorder: worldwide experience," *Mol Psychiatry* 15 (1) (2010), pp. 64-79. cited by applicant

Greenberg. et al., "Three-year outcomes in deep brain stimulation for highly resistant obsessive-compulsive disorder," *Neuropsychopharmacology* 31 (11) (2006), pp. 2384-2393. cited by

applicant

Gutman, et al., "A tractography analysis of two deep brain stimulation white matter targets for depression," *Biol Psychiatry* 65 (4) (2009), pp. 276-282. cited by applicant

Haber, et al., "Reward-related cortical inputs define a large striatal region in primates that interface with associative cortical connections, providing a substrate for incentive-based learning," *J Neurosci* 26 (32) (2006), pp. 8368-8376. cited by applicant

Haber, et al., "Cognitive and limbic circuits that are affected by deep brain stimulation," *Front Biosci* 14 (2009), pp. 1823-1834. cited by applicant

Hines, M. L., et al., "The NEURON simulation environment," *Neural Comput.*, 9(6) (Aug. 15, 1997), pp. 1179-1209. cited by applicant

Hua, et al., "Tract probability maps in stereotaxic spaces: analyses of white matter anatomy and tract-specific quantification," *Neuroimage* 39 (1) (2008), pp. 336-347. cited by applicant

Johansen-Berg, et al., "Anatomical connectivity of the subgenual cingulate region targeted with deep brain stimulation for treatment-resistant depression," *Cereb Cortex* 18 (6) (2008), pp. 1374-1383. cited by applicant

Kopell, et al., "Deep brain stimulation for psychiatric disorders," *J Clin Neurophysiol* 21 (1) (2004), pp. 51-67. cited by applicant

Lozano, et al., "Subcallosal cingulate gyrus deep brain stimulation for treatment-resistant depression," *Biol Psychiatry* 64 (6) (2008), pp. 461-467. cited by applicant

Lujan, et al., "Tracking the mechanisms of deep brain stimulation for neuropsychiatric disorders," *Front Biosci* 13 (2008), pp. 5892-5904. cited by applicant

Lujan, J.L. et al., "Automated 3-Dimensional Brain Atlas Fitting to Microelectrode Recordings from Deep Brain Stimulation Surgeries," *Stereotact. Funct. Neurosurg.* 87(2009), pp. 229-240. cited by applicant

Machado. et al., "Functional topography of the ventral striatum and anterior limb of the internal capsule determined by electrical stimulation of awake patients," *Clin Neurophysiol* 120 (11) (2009), pp. 1941-1948. cited by applicant

Malone, et al., "Deep brain stimulation of the ventral capsule/ventral striatum for treatment-resistant depression," *Biol Psychiatry* 65 (4) (2009), pp. 267-275. cited by applicant

Mayberg, H. S., et al., "Deep brain stimulation for treatment-resistant depression," *Neuron*, 45(5) (Mar. 3, 2005), pp. 651-660. cited by applicant

Mayberg, H. S., et al., "Limbic-cortical dysregulation: a proposed model of depression," *J Neuropsychiatry Clin Neurosci.* 9 (3) (1997), pp. 471-481. cited by applicant

McIntyre, C. C., et al., "Network perspectives on the mechanisms of deep brain stimulation," *Neurobiol Dis* 38 (3) (2010), pp. 329-337. cited by applicant

Miocinovic, S., et al., "Experimental and theoretical characterization of the voltage distribution generated by deep brain stimulation," *Exp Neurol* 216 (i) (2009), pp. 166-176. cited by applicant

Nuttin, et al., "Electrical stimulation in anterior limbs of internal capsules in patients with obsessive-compulsive disorder," *Lancet* 354 (9189) (1999), p. 1526. cited by applicant

Saxena, et al., "Cerebral glucose metabolism in obsessive-compulsive hoarding," *Am J Psychiatry.* 161 (6) (2004), pp. 1038-1048. cited by applicant

Viola, et al., "Importance-driven focus of attention," *IEEE Trans Vis Comput Graph* 12 (5) (2006), pp. 933-940. cited by applicant

Wakana, S., et al., "Reproducibility of quantitative tractography methods applied to cerebral white matter," *Neuroimage* 36 (3) (2007), pp. 630-644. cited by applicant

Mayr et al., "Basic Design and Construction of the Vienna FES Implants: Existing Solutions and Prospects for New Generations of Implants", *Medical Engineering & Physics*, 2001; 23:53-60. cited by applicant

Nowinski, W. L., et al., "Statistical analysis of 168 bilateral subthalamic nucleus implantations by means of the probabilistic functional atlas.", *Neurosurgery* 57(4 Suppl) (Oct. 2005), 319-30. cited

by applicant

Obeso, J. A., et al., "Deep-brain stimulation of the subthalamic nucleus or the pars interna of the globus pallidus in Parkinson's disease.", *N Engl J Med.*, 345{131. The Deep-Brain Stimulation for Parkinson's Disease Study Group, (Sep. 27, 2001),956-63. cited by applicant

Butson et al.. "Current Steering to control the volume of tissue activated during deep brain stimulation," vol. 1, No. 1, Dec. 3, 2007, pp. 7-15. cited by applicant

Patrick, S. K., et al., "Quantification of the UPDRS rigidity scale", *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, [see also *IEEE Trans. on Rehabilitation Engineering* 9(1). (2001),31-41. cited by applicant

Phillips, M. D., et al., "Parkinson disease: pattern of functional MR imaging activation during deep brain stimulation of subthalamic nucleus—initial experience", *Radiology* 239(1). (Apr. 2006), 209-16. cited by applicant

Ericsson, A. et al., "Construction of a patient-specific atlas of the brain: Application to normal aging," *Biomedical Imaging: From Nano to Macro*, ISBI 2008, 5th IEEE International Symposium, May 14, 2008, pp. 480-483. cited by applicant

Kaikai Shen et al., "Atlas selection strategy using least angle regression in multi-atlas segmentation propagation," *Biomedical Imaging: From Nano to Macro*, 2011, 8th IEEE International Symposium, ISBI 2011, Mar. 30, 2011, pp. 1746-1749. cited by applicant

Liliane Ramus et al., "Assessing selection methods in the cotnext of multi-atlas based segmentation," *Biomedical Imaging: From Nano to Macro*, 2010, IEEE International Symposium, Apr. 14, 2010, pp. 1321-1324. cited by applicant

Olivier Commowick et al., "Using Frankenstein's Creature Paradigm to Build a Patient Specific Atlas," Sep. 20, 2009, *Medical Image Computing and Computer-Assisted Intervention*, pp. 993-1000. cited by applicant

Lotjonen J.M.P. et al., "Fast and robust multi-atlas segmentation of brain magnetic resonance images," *NeuroImage*, Academic Press, vol. 49, No. 3, Feb. 1, 2010, pp. 2352-2365. cited by applicant

McIntyre, C. C., et al., "How does deep brain stimulation work? Present understanding and future questions.", *J Clin Neurophysiol.* 21 (1). (Jan.-Feb. 2004), 40-50. cited by applicant

Sanchez Castro et al., "A cross validation study of deep brain stimulation targeting: From experts to Atlas-Based, Segmentation-Based and Automatic Registration Algorithms," *IEEE Transactions on Medical Imaging*, vol. 25, No. 11, Nov. 1, 2006, pp. 1440-1450. cited by applicant

Plaha, P. , et al., "Stimulation of the caudal zona incerta is superior to stimulation of the subthalamic nucleus in Improving contralateral parkinsonism.", *Brain* 129{Pt 7} (Jul. 2006), 1732-47. cited by applicant

Rattay, F, "Analysis of models for external stimulation of axons", *IEEE Trans. Biomed. Eng.* vol. 33 (1986), 974-977. cited by applicant

Rattay, F., "Analysis of the electrical excitation of CNS neurons", *IEEE Transactions on Biomedical Engineering* 45 (6). (Jun. 1998), 766-772. cited by applicant

Rose, T. L., et al., "Electrical stimulation with Pt electrodes. VIII. Electrochemically safe charge injection limits with 0.2 ms pulses [neuronal application]", *IEEE Transactions on Biomedical Engineering*, 37(11 }, (Nov. 1990), 1118-1120. cited by applicant

Rubinstein, J. T., et al., "Signal coding in cochlear implants: exploiting stochastic effects of electrical stimulation", *Ann Otol Rhinol Laryngol Suppl* . . . 191, (Sep. 2003), 14-9. cited by applicant

Schwan, H.P., et al., "The conductivity of living tissues.", *Ann NY Acad Sci.*, 65(6). (Aug., 1957), 1007-13. cited by applicant

Taylor, R. S., et al., "Spinal cord stimulation for chronic back and leg pain and failed back surgery syndrome: a systematic review and analysis of prognostic factors", *Spine* 30(1). (Jan. 1, 2005), 152-60. cited by applicant

Siegel, Ralph M. et al., "Spatiotemporal dynamics of the functional architecture for gain fields in inferior parietal obule of behaving monkey," *Cerebral Cortex*, New York, NY, vol. 17, No. 2, Feb. 2007, pp. 378-390. cited by applicant

Klein, A. et al., "Evaluation of 14 nonlinear deformation algorithms applied to human brain MRI registration," *NeuroImage*, Academic Press, Orlando, FL, vol. 46, No. 3, Jul. 2009, pp. 786-802. cited by applicant

Geddes, L. A., et al., "The specific resistance of biological material-a compendium of data for the biomedical engineer and physiologist.," *Med Biol Eng*. 5(3). (May 1967),271-93. cited by applicant

Gimsa, J., et al., "Choosing electrodes for deep brain stimulation experiments-electrochemical considerations.," *J Neurosci Methods*, 142(2), (Mar. 30, 2005),251-65. cited by applicant

Vidailhet, M. , et al., "Bilateral deep-brain stimulation of the globus pallidus in primary generalized dystonia", *N Engl J Med*. 352(5) (Feb. 3, 2005),459-67. cited by applicant

Izad, Oliver, "Computationally Efficient Method in Predicating Axonal Excitation," Dissertation for Master Degree, Department of Biomedical Engineering, Case Western Reserve University, May 2009. cited by applicant

Jaccard, Paul, "Elude comparative de la distribution florale dans une portion odes Aples et des Jura," *Bulletin de la Societe Vaudoise des Sciences Naturelles* (1901), 37:547-579. cited by applicant

Dice, Lee R., "Measures of the Amount of Ecologic Association Between Species," *Ecology* 26(3) (1945): 297-302. doi: 10.2307/ 1932409, <http://jstor.org/stable/1932409>. cited by applicant

Rand, WM., "Objective criteria for the evaluation of clustering methods," *Journal of the American Statistical Association* (American Statistical Association) 66 (336) (1971): 846-850, doi:10.2307/2284239, <http://jstor.org/stable/2284239>. cited by applicant

Hubert, Lawrence et al., "Comparing partitions," *Journal of Classification* 2(1) (1985): 193-218, doi:10.1007/ BF01908075. cited by applicant

Cover, T.M. et al., "Elements of information theory," (1991) John Wiley & Sons, New York, NY. cited by applicant

Meila, Marina, "Comparing Clusterings by the Variation of Information," *Learning Theory and Kernel Machines* (2003): 173-187. cited by applicant

Viola, P., et al., "Alignment by maximization of mutual information", *International Journal of Computer Vision* 24(2). (1997), 137-154. cited by applicant

Butson et al. "StimExplorer: Deep Brain Stimulation Parameter Selection Software System," *Acta Neurochirurgica*, Jan. 1, 2007, vol. 97, No. 2, pp. 569-574. cited by applicant

Butson et al. "Role of Electrode Design on the vol. of Tissue Activated During Deep Brain Stimulation," *Journal of Neural Engineering*, Mar. 1, 2006, vol. 3, No. 1, pp. 1-8. cited by applicant

Volkman et al., *Introduction to the Programming of Deep Brain Stimulators*, *Movement Disorders*, vol. 17, Suppl. 3, pp. S181-S187 (2002). cited by applicant

Miocinovic et al. "Cicerone: Stereotactic Neurophysiological Recording and Deep Brain Stimulation Electrode Placement Software System," *Acta Neurochirurgica Suppl.*, Jan. 1, 2007, vol. 97, No. 2, pp. 561-567. cited by applicant

Schmidt et al. "Sketching and Composing Widgets for 3D Manipulation," *Eurographics*, Apr. 2008, vol. 27, No. 2, pp. 301-310. cited by applicant

Volkman, J. , et al., "Basic algorithms for the programming of deep brain stimulation in Parkinson's disease", *Mov Disord.*, 21 Suppl 14. (Jun. 2006),S284-9. cited by applicant

Walter, B. L., et al., "Surgical treatment for Parkinson's disease", *Lancet Neural*. 3(12). (Dec. 2004),719-28. cited by applicant

Wei, X. F., et al., "Current density distributions, field distributions and impedance analysis of segmented deep brain stimulation electrodes", *J Neural Eng* . . . 2(4). (Dec. 2005), 139-47. cited by applicant

Zonenshayn, M. , et al., "Location of the active contact within the subthalamic nucleus (STN) in

the treatment of diopathic Parkinson's disease.”, *Surg Neurol.*, 62(3) (Sep. 2004), 216-25. cited by applicant

Da Silva et al (A primer on diffusion tensor imaging of anatomical substructures. *Neurosurg Focus* 15(1): p. 1-4, Article 4, 2003.). cited by applicant

Micheli-Tzanakou, E., et al., “Computational Intelligence for target assesment in Parkinson's disease”, *Proceedings of SPIE vol. 4479. Applications and Science of Neural Networks, Fuzzy Systems, and Evolutionary Computation IV*,(2001),54-69. cited by applicant

Grill, W. M., “Stimulus waveforms for selective neural stimulation”, *IEEE Engineering in Medicine and Biology Magazine*, 14(4), (Jul.-Aug. 1995), 375-385. cited by applicant

Miocinovic, S., et al., “Sensitivity of temporal excitation properties to the neuronal element activated by extracellular stimulation”, *J Neurosci Methods*. 132(1). (Jan. 15, 2004), 91-9. cited by applicant

Hunka, K. et al., *Nursing Time to Program and Assess Deep Brain Stimulators in Movement Disorder Patients*, *J. Neursci Nurs.*, 37: 204-10 (Aug. 2005). cited by applicant

Moss, J. , et al., “Electron microscopy of tissue adherent to explanted electrodes in dystonia and Parkinson's disease”, *Brain*, 127{Pt 12}. (Dec., 2004),2755-63. cited by applicant

Montgomery, E. B., et al., “Mechanisms of deep brain stimulation and future technical developments.”, *Neurol Res*. 22(3). (Apr. 2000),259-66. cited by applicant

Merrill, D. R., et al., “Electrical stimulation of excitable tissue: design of efficacious and safe protocols”, *J Neurosci Methods*. 141(2), (Feb. 15, 2005), 171-98. cited by applicant

Fisekovic et al., “New Controller for Functional Electrical Stimulation Systems”, *Med. Eng. Phys.* 2001; 23:391-399. cited by applicant

McIntyre, Cameron , et al., “Finite element analysis of the current-density and electric field generated by metal microelectrodes”, *Ann Biomed Eng .* 29(3), (2001),227-235. cited by applicant

Foster, K. R., et al., “Dielectric properties of tissues and biological materials: a critical review.”, *Grit Rev Biomed Ena*. 17(1). {1989},25-104. cited by applicant

Limousin, P., et al., “Electrical stimulation of the subthalamic nucleus in advanced Parkinson's disease”, *N Engl J Med . . .* 339(16), (Oct. 15, 1998), 1105-11. cited by applicant

Kitagawa, M., et al., “Two-year follow-up of chronic stimulation of the posterior subthalamic white matter for tremor-dominant Parkinson's disease.”, *Neurosurgery*. 56(2). (Feb. 2005),281-9. cited by applicant

Johnson, M. D., et al., “Repeated voltage biasing improves unit recordings by reducing resistive tissue impedances”, *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, [see also *IEEE Trans. on Rehabilitation Engineering* (2005), 160-165. cited by applicant

Holsheimer, J. , et al., “Chronaxie calculated from current-duration and voltage-duration data”, *J Neurosci Methods*. 97(1). (Apr. 1, 2000),45-50. cited by applicant

Hines, M. L., et al., “The NEURON simulation environment”, *Neural Comput*. 9(6). (Aug. 15, 1997), 1179-209. cited by applicant

Herzog, J., et al., “Most effective stimulation site in subthalamic deep brain stimulation for Parkinson's disease”, *Mov Disord*. 19(9). (Sep. 2004),1050-4. cited by applicant

Hershey, T., et al., “Cortical and subcortical blood flow effects of subthalamic nucleus stimulation in PD.”, *Neurology* 61(6). (Sep. 23, 2003),816-21. cited by applicant

Hemm, S. , et al., “Evolution of Brain Impedance in Dystonic Patients Treated by GPi Electrical Stimulation”, *Neuromodulation* 7(2) (Apr. 2004),67-75. cited by applicant

Jemm, S., et al., “Deep brain stimulation in movement disorders: stereotactic coregistration of two-dimensional electrical field modeling and magnetic resonance imaging.”, *J Neurosurg*. 103(6): (Dec. 2005),949-55. cited by applicant

Haueisen, J, et al., “The influence of brain tissue anisotropy on human EEG and MEG”, *Neuroimage* 15(1) (Jan. 2002),159-166. cited by applicant

Haslinger, B., et al., “Frequency-correlated decreases of motor cortex activity associated with

subthalamic nucleus stimulation in Parkinson's disease.”, *Neuroimage* 28(3). (Nov. 15, 2005),598-606. cited by applicant

Hashimoto, T. , et al., “Stimulation of the subthalamic nucleus changes the firing pattern of pallidal neurons”, *J Neurosci.* 23(5). (Mar. 1, 2003),1916-23. cited by applicant

Hardman, C. D., et al., “Comparison of the basal ganglia in rats, marmosets, macaques, baboons, and humans: vol. and neuronal number for the output, internal relay, and striatal modulating nuclei”, *J Comp Neurol.*, 445(3). (Apr. 8, 2002),238-55. cited by applicant

McNaughtan et al., “Electrochemical Issues in Impedance Tomography”, 1st World Congress on Industrial Process Tomography, Buxton, Greater Manchester, Apr. 14-17, 1999. cited by applicant

Grill, WM., et al., “Electrical properties of implant encapsulation tissue”, *Ann Biomed Eng.* vol. 22. (1994),23-33. cited by applicant

Grill, W. M., et al., “Deep brain stimulation creates an informational lesion of the stimulated nucleus”, *Neuroreport.* 1517t (May 19, 2004), 1137-40. cited by applicant

International Search Report and Written Opinion for PCT Application No. PCT/US2022/026162 mailed Jul. 25, 2022. cited by applicant

Pulliam CL, Heldman DA, Orcutt TH, Mera TO, Giuffrida JP, Vitek JL. Motion sensor strategies for automated optimization of deep brain stimulation in Parkinson's disease. *Parkinsonism Relat Disord.* Apr. 2015;21(4):378-82. cited by applicant

Benoit M. Dawant et al: “The VU-DBS project: integrated and computer-assisted planning, intra-operative placement, and post-operative programming of deep-brain stimulators”, *Proceedings of SPIE*, vol. 6509, Mar. 6, 2007 (Mar. 6, 2007), 11 pages. cited by applicant

Mitra PP, Pesaran B. Analysis of dynamic brain imaging data. *Biophys J.* Feb. 1999; 76(2):691-708. doi: 10.1016/S0006-3495(99)77236-X. PMID: 9929474; PMCID: PMC1300074. cited by applicant

Hammer N, Glätzner J, Feja C, Kühne C, Meixensberger J, et al. (2015) Human Vagus Nerve Branching in the Cervical Region. *Plos One* 10(2): e0118006. Published: Feb. 13, 2015. <https://doi.org/10.1371/journal.pone.0118006>. cited by applicant

Trost M, Su S, Su P, Yen RF, Tseng HM, Barnes A, Ma Y, Eidelberg D. Network modulation by the subthalamic nucleus in the treatment of Parkinson's disease. *Neuroimage.* May 15, 2006;31(1):301-7. doi: 10.1016/j.neuroimage.2005.12.024. Epub Feb. 8, 2006. cited by applicant

Mcintyre, Cameron C., et al., “Electric Field and Stimulating Influence generated by Deep Brain Stimulation of the Subthalamaic Nucleus,” *Clinical Neurophysiology*, 115(3) (Mar. 2004), pp. 589-595. cited by applicant

Mcintyre, Cameron C., et al., “Electric field generated by deep brain stimulation of the subthalamic nucleus,” *Biomedical Engineering Society Annual Meeting*, Nashville TN (Oct. 2003), 16 pages. cited by applicant

Mcintyre, Cameron C., et al., “Excitation of central nervous system neurons by nonuniform electric fields,” *Biophys. J.*, 76(2) (1999), pp. 878-888. cited by applicant

McNeal, DR., et al., “Analysis of a model for excitation of myelinated nerve,” *IEEE Trans Biomed Eng.*, vol. 23 (1976), pp. 329-337. cited by applicant

Micheli-Tzanakou, E. , et al., “Computational Intelligence for target assesment in Parkinson's disease,” *Proceedings of SPIE* vol. 4479, *Applications and Science of Neural Networks, Fuzzy Systems, and Evolutionary Computation IV* (2001), pp. 54-69. cited by applicant

Miocinovic, S., et al., “Computational analysis of subthalamic nucleus and lenticular fasciculus activation during therapeutic deep brain stimulation,” *J Neurophysiol.*, 96(3) (Sep. 2006), pp. 1569-1580. cited by applicant

Miranda, P. C., et al., “The distribution of currents induced in the brain by Magnetic Stimulation: a finite element analysis incorporating OT-MRI-derived conductivity data,” *Proc. Intl. Soc. Mag. Reson. Med.* 9 (2001), p. 1540. cited by applicant

Miranda, P. C., et al., “The Electric Field Induced in the Brain by Magnetic Stimulation: A 3-D

Finite-Element Analysis of the Effect of Tissue Heterogeneity and Anisotropy," IEEE Transactions on Biomedical Engineering, 50(9) (Sep. 2003), pp. 1074-1085. cited by applicant

Moffitt, MA., et al., "Prediction of myelinated nerve fiber stimulation thresholds: limitations of linear models," IEEE Transactions on Biomedical Engineering, 51 (2) (2003), pp. 229-236. cited by applicant

Moro, E, et al., "The impact on Parkinson's disease of electrical parameter settings in STN stimulation," Neurology, 59 (5) (Sep. 10, 2002), pp. 706-713. cited by applicant

Nowak, LG., et al., "Axons, but not cell bodies, are activated by electrical stimulation in cortical gray matter. I. Evidence from chronaxie measurements," Exp. Brain Res., 118(4) (Feb. 1998), pp. 477-488. cited by applicant

Nowak, LG., et al., "Axons, but not cell bodies, are activated by electrical stimulation in cortical gray matter. II. Evidence from selective inactivation of cell bodies and axon initial segments," Exp. Brain Res., 118(4) (Feb. 1998), pp. 489-500. cited by applicant

O'Suilleabhain, PE., et al., "Tremor response to polarity, voltage, pulsewidth and frequency of thalamic stimulation," Neurology, 60(5) (Mar. 11, 2003), pp. 786-790. cited by applicant

Pierpaoli, C., et al., "Toward a quantitative assessment of diffusion anisotropy," Magn Reson Med., 36(6) (Dec. 1996), pp. 893-906. cited by applicant

Plonsey, R., et al., "Considerations of quasi-stationarity in electrophysiological systems," Bull Math Biophys., 29(4) (Dec. 1967), pp. 657-664. cited by applicant

Ranck, J B., "Specific impedance of rabbit cerebral cortex," Exp. Neurol., vol. 7 (Feb. 1963), pp. 144-152. cited by applicant

Ranck, J B., et al., "The Specific impedance of the dorsal columns of the cat: an anisotropic medium," Exp. Neurol., 11 (Apr. 1965), pp. 451-463. cited by applicant

Ranck, J B., "Which elements are excited in electrical stimulation of mammalian central nervous system: a review," Brain Res., 98(3) (Nov. 21, 1975), pp. 417-440. cited by applicant

Rattay, F., et al., "A model of the electrically excited human cochlear neuron. I. Contribution of neural substructures to the generation and propagation of spikes," Hear Res., 153(1-2) (Mar. 2001), pp. 43-63. cited by applicant

Rattay, F., "A model of the electrically excited human cochlear neuron. II. Influence of the three-dimensional cochlear structure on neural excitability," Hear Res., 153(1-2) (Mar. 2001), pp. 64-79. cited by applicant

Rattay, F., "Arrival at Functional Electrostimulation by modelling of fiber excitation," Proceedings of the Ninth annual Conference of the IEEE Engineering in Medicine and Biology Society (1987), pp. 1459-1460. cited by applicant

Rattay, F., "The influence of intrinsic noise can preserve the temporal fine structure of speech signals in models of electrically stimulated human cochlear neurones," Journal of Physiology, Scientific Meeting of the Physiological Society, London, England, UK Apr. 19-21, 1999 (Jul. 1999), p. 170P. cited by applicant

Rizzone, M., et al., "Deep brain stimulation of the subthalamic nucleus in Parkinson's disease: effects of variation in stimulation parameters," J. Neurol. Neurosurg. Psychiatry., 71(2) (Aug. 2001), pp. 215-219. cited by applicant

Saint-Cyr, J. A., et al., "Localization of clinically effective stimulating electrodes in the human subthalamic nucleus on magnetic resonance imaging," J. Neurosurg., 87(5) (Nov. 2002), pp. 1152-1166. cited by applicant

Sances, A., et al., "In Electroanesthesia: Biomedical and Biophysical Studies," A Sances and SJ Larson, Eds., Academic Press, NY (1975), pp. 114-124. cited by applicant

Sl. Jean, P., et al., "Automated atlas integration and interactive three-dimensional visualization tools for planning and guidance in functional neurosurgery," IEEE Transactions on Medical Imaging, 17(5) (1998), pp. 672-680. cited by applicant

Starr, P.A., et al., "Implantation of deep brain stimulators into the subthalamic nucleus: technical

approach and magnetic resonance imaging-verified lead locations,” J. Neurosurg., 97(2) (Aug. 2002), pp. 370-387. cited by applicant

Sterio, D., et al., “Neurophysiological refinement of subthalamic nucleus targeting,” Neurosurgery, 50(1) (Jan. 2002), pp. 58-69. cited by applicant

Struijk, J. J., et al., “Excitation of dorsal root fibers in spinal cord stimulation: a theoretical study,” IEEE Transactions on Biomedical Engineering, 40(7) (Jul. 1993), pp. 632-639. cited by applicant

Struijk, J. J., et al., “Recruitment of dorsal col. fibers in spinal cord stimulation: influence of collateral branching,” IEEE Transactions on Biomedical Engineering, 39(9) (Sep. 1992), pp. 903-912. cited by applicant

Tamma, F., et al., “Anatomo-clinical correlation of intraoperative stimulation-induced side effects during HF-DBS of the subthalamic nucleus,” Neurol Sci., vol. 23 (Suppl 2) (2002), pp. 109-110. cited by applicant

Tarler, M., et al., “Comparison between monopolar and tripolar configurations in chronically implanted nerve cuff electrodes,” IEEE 17th Annual Conference Engineering in Medicine and Biology Society, vol. 2 (1995), pp. 1093-1109. cited by applicant

Testerman, Roy L., “Coritcal response to callosal stimulation: A model for determining safe and efficient stimulus parameters,” Annals of Biomedical Engineering, 6(4) (1978), pp. 438-452. cited by applicant

Tuch, D.S., et al., “Conductivity mapping of biological tissue using diffusion MRI,” Ann NY Acad Sci., 888 (Oct. 30, 1999), pp. 314-316. cited by applicant

Tuch, D.S., et al., “Conductivity tensor mapping of the human brain using diffusion tensor MRI,” Proc Nall Acad Sci USA, 98(20) (Sep. 25, 2001), pp. 11697-11701. cited by applicant

Veraart, C., et al., “Selective control of muscle activation with a multipolar nerve cuff electrode,” IEEE Transactions on Biomedical Engineering, 40(7) (Jul. 1993), pp. 640-653. cited by applicant

Vercueil, L., et al., “Deep brain stimulation in the treatment of severe dystonia,” J. Neurol., 248(8) (Aug. 2001), pp. 695-700. cited by applicant

Vilalte, “Circuit Design of the Power-on-Reset,” Apr. 2000, pp. 1-25. cited by applicant

Vitek, J. L., “Mechanisms of deep brain stimulation: excitation or inhibition,” Mov. Disord., vol. 17 (Suppl. 3) (2002), pp. 69-72. cited by applicant

Voges, J., et al., “Bilateral high-frequency stimulation in the subthalamic nucleus for the treatment of Parkinson disease: correlation of therapeutic effect with anatomical electrode position,” J. Neurosurg., 96(2) (Feb. 2002), pp. 269-279. cited by applicant

Wakana, S., et al., “Fiber tract-based atlas of human white matter anatomy,” Radiology, 230(1) (Jan. 2004), pp. 77-87. cited by applicant

Alexander, DC., et al., “Spatial transformations of diffusion tensor magnetic resonance images,” IEEE Transactions on Medical Imaging, 20 (11), (2001), pp. 1131-1139. cited by applicant

Wu, Y. R., et al., “Does Stimulation of the GPi control dyskinesia by activating inhibitory axons?,” Mov. Disord., vol. 16 (2001), pp. 208-216. cited by applicant

Yelnik, J., et al., “Localization of stimulating electrodes in patients with Parkinson disease by using a three-dimensional atlas-magnetic resonance imaging coregistration method,” J Neurosurg., 99(1) (Jul. 2003), pp. 89-99. cited by applicant

Yianni, John, et al., “Globus pallidus internus deep brain stimulation for dystonic conditions: a prospective audit,” Mov. Disord., vol. 18 (2003), pp. 436-442. cited by applicant

Zonenshayn, M., et al., “Comparison of anatomic and neurophysiological methods for subthalamic nucleus targeting,” Neurosurgery, 47(2) (Aug. 2000), pp. 282-294. cited by applicant

Voghell et al., “Programmable Current Source Dedicated to Implantable Microstimulators” ICM '98 Proceedings of the Tenth International Conference, pp. 67-70, (Dec. 1998). cited by applicant

Butson, Christopher R., et al., “Patient-specific analysis of the volume of tissue activated during deep brain stimulation”, Neurolmage. vol. 34. (2007),661-670. cited by applicant

Adler, DE., et al., “The tentorial notch: anatomical variation, morphometric analysis, and

classification in 100 human autopsy cases,” J. Neurosurg., 96(6), (Jun. 2002), pp. 1103-1112. cited by applicant
Jones et al., “An Advanced Demultiplexing System for Physiological Stimulation”, IEEE Transactions on Biomedical Engineering, vol. 44 No. Dec. 12, 1997, pp. 1210-1220. cited by applicant

Primary Examiner: Patel; Niketa

Assistant Examiner: Hussaini; Attiya Sayyada

Attorney, Agent or Firm: Branch Partners PLLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit under 35 U.S.C. § 119(e) of U.S. Provisional Patent Application Ser. No. 63/180,380, filed Apr. 27, 2021, which is incorporated herein by reference.

FIELD

(1) The present disclosure is directed to the area of methods and systems for at least partially automating programming of electrical stimulation. The present disclosure is also directed to methods and systems for automating the testing of multiple sets of electrical stimulation parameters.

BACKGROUND

(2) Implantable electrical stimulation systems have proven therapeutic in a variety of diseases and disorders. For example, deep brain stimulation systems have been used as a therapeutic modality for the treatment of Parkinson's disease, essential tremor, and the like.

(3) Stimulators have been developed to provide therapy for a variety of treatments. A stimulator can include an implantable pulse generator (IPG), one or more leads, and an array of stimulator electrodes on each lead. The stimulator electrodes are in contact with or near the nerves, muscles, or other tissue to be stimulated. The pulse generator in the IPG generates electrical pulses that are delivered by the electrodes to body tissue.

(4) Implantable medical devices (IMDs), including IPGs, typically have the capability to communicate data with an external device, such as a clinician programmer or a remote control, via a radio-frequency telemetry link or other wireless communication method. The clinician programmer can program the operating parameters of the implanted medical device. The remote control can switch programs. Modern implantable devices also include the capability for bidirectional communication so that information can be transmitted to the clinician programmer or remote control from the implanted device.

BRIEF SUMMARY

(5) One aspect is a method for automating selection of stimulation parameters for a stimulation device implanted in a patient. The method includes setting, by a user, at least one limit on each of at least one stimulation parameter and performing, automatically using at least one processor, the following actions for each of a plurality of sets of the stimulation parameters constrained by the at least one limit: stimulating the patient, by the stimulation device, using the set of stimulation parameters, sensing one or more effects arising in response to the stimulation, and updating, by the at least one processor, a collection of the effects and sets of stimulation parameters with the one or more effects and the set of stimulation parameters. The method further includes selecting, by the processor, one of the sets of stimulation parameters based on the effects.

(6) In at least some aspects, the method further includes, after selecting the one of the sets of stimulation parameters, fine-tuning, manually by the user, the stimulation parameters of the one of the sets of stimulation parameters. In at least some aspects, the method further includes programming the stimulation device with the one of the sets of stimulation parameters.

(7) In at least some aspects, the sensing includes sensing a signal from the stimulated tissue. In at least some aspects, the signal is an evoked compound action potential (ECAP), evoked resonant neural activity (ERNA), or local field potential (LFP). In at least some aspects, at least one of the one or more effects is a feature of the signal, wherein the feature is a range of the signal, an area under a curve of the signal, a length of the curve of the signal, a maximum of the signal, or a minimum of the signal.

(8) In at least some aspects, the performing includes performing the actions for sets of stimulation parameters at different axial locations along an electrical stimulation lead and selecting a one of the different axial locations based on the sensed one or more effects. In at least some aspects, the performing further includes, after selecting the one of the different axial locations, performing the actions for sets of stimulation parameters at second axial locations within one axial electrode spacing of the selected one of the different axial locations and selecting one of the second axial locations or the selected one of the different axial locations. In at least some aspects, the performing further includes, after selecting the one of the different axial locations, performing the actions for sets of stimulation parameters at different rotational directions around the electrical stimulation lead at the selected one of the different axial locations and selecting a one of the different rotational directions based on the sensed one or more effects. In at least some aspects, the performing further includes, after selecting the one of the different rotational directions, performing the actions for sets of stimulation parameters at second rotational directions within 120 degrees of the selected one of the different rotational directions and selecting one of the second rotational directions or the selected one of the different rotational directions. In at least some aspects, the performing further includes, after selecting the one of the different rotational directions, performing the actions for sets of stimulation parameters at different stimulation amplitudes for the selected one of the different rotational directions and selecting one of the stimulation amplitudes.

(9) Another aspect is a system for selection of stimulation parameters for a stimulation device implanted in a patient. The system includes at least one processor configured to perform actions, the actions including receiving, from a user, at least one limit for each of at least one stimulation parameter and performing the following actions for each of a plurality of sets of the stimulation parameters constrained by the at least one limit: directing the stimulation device to stimulate the patient using the set of stimulation parameters, sensing one or more effects arising in response to the stimulation, and updating a collection of the effects and sets of stimulation parameters with the one or more effects and the set of stimulation parameters. The actions further include selecting one of the sets of stimulation parameters based on the effects.

(10) In at least some aspects, the system further includes the stimulation device. In at least some aspects, the stimulation device includes an electrical stimulation lead having electrodes disposed along a distal portion of the electrical stimulation lead. In at least some aspects, sensing one or more effects including sensing the one or more effects arising in response to the stimulation using at least one of the electrodes of the electrical stimulation lead. In at least some aspects, the system further includes a sensor external to the stimulation device, wherein sensing one or more effects including sensing the one or more effects arising in response to the stimulation using the sensor. In at least some aspects, the actions further include programming the stimulation device with the one of the sets of stimulation parameters.

(11) Yet another aspect is a non-transitory computer-readable medium having stored thereon instructions for execution by at least one processor. The instructions include receiving, from a user, at least one limit for each of at least one stimulation parameter and performing the following actions for each of a plurality of sets of the stimulation parameters constrained by the at least one

limit: directing the stimulation device to stimulate the patient using the set of stimulation parameters, sensing one or more effects arising in response to the stimulation, and updating a collection of the effects and sets of stimulation parameters with the one or more effects and the set of stimulation parameters. The instructions further including selecting one of the sets of stimulation parameters based on the effects.

(12) In at least some aspects, the instructions further include, after selecting the one of the sets of stimulation parameters, fine-tuning, manually by the user, the stimulation parameters of the one of the sets of stimulation parameters. In at least some aspects, the instructions further include programming the stimulation device with the one of the sets of stimulation parameters. In at least some aspects, the instructions further include repeating the performing instruction periodically to track disease progression over time

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Non-limiting and non-exhaustive embodiments of the present invention are described with reference to the following drawings. In the drawings, like reference numerals refer to like parts throughout the various figures unless otherwise specified.

(2) For a better understanding of the present invention, reference will be made to the following Detailed Description, which is to be read in association with the accompanying drawings, wherein:

(3) FIG. 1 is a schematic view of one embodiment of an electrical stimulation system that includes one or more leads that can be coupled to an IPG;

(4) FIG. 2 is a block diagram of elements of an electrical stimulation system;

(5) FIG. 3A is a schematic perspective view of a distal portion of one embodiment of an electrical stimulation lead with segmented electrodes;

(6) FIG. 3B is a schematic perspective view of a distal portion of another embodiment of an electrical stimulation lead with segmented electrodes;

(7) FIG. 3C is a schematic perspective view of a distal portion of a third embodiment of an electrical stimulation lead with segmented electrodes;

(8) FIG. 3D is a schematic perspective view of a distal portion of a fourth embodiment of an electrical stimulation lead with segmented electrodes;

(9) FIG. 3E is a schematic perspective view of a distal portion of a fifth embodiment of an electrical stimulation lead with segmented electrodes;

(10) FIG. 4 is a flowchart of one embodiment of a method of determining stimulation parameters for therapy;

(11) FIG. 5 is a flowchart of one embodiment of a method for selection of a set of stimulation parameters;

(12) FIG. 6 is a flowchart of one embodiment of a method of sensing a stimulation effect;

(13) FIG. 7 is a schematic illustration of one embodiment of a user interface with a clinical effects map; and

(14) FIG. 8 is schematic illustration of one embodiment of a signal measured using a sensor in response to stimulation.

DETAILED DESCRIPTION

(15) The present disclosure is directed to the area of methods and systems for at least partially automating programming of electrical stimulation. The present disclosure is also directed to methods and systems for automating the testing of multiple sets of electrical stimulation parameters.

(16) Implantable electrical stimulation systems and devices are used herein to exemplify the inventions, but it will be understood that these inventions can be utilized with other stimulation

systems and devices. Examples of implantable electrical stimulation systems include, but are not limited to, a least one lead with one or more electrodes disposed along a distal end of the lead and one or more terminals disposed along the one or more proximal ends of the lead. Examples of electrical stimulation systems with leads are found in, for example, U.S. Pat. Nos. 6,181,969; 6,295,944; 6,391,985; 6,516,227; 6,609,029; 6,609,032; 6,741,892; 7,244,150; 7,450,997; 7,672,734; 7,761,165; 7,783,359; 7,792,590; 7,809,446; 7,949,395; 7,974,706; 8,831,742; 8,688,235; 8,175,710; 8,224,450; 8,271,094; 8,295,944; 8,364,278; and 8,391,985; U.S. Patent Application Publications Nos. 2007/0150036; 2009/0187222; 2009/0276021; 2010/0076535; 2010/0268298; 2011/0004267; 2011/0078900; 2011/0130817; 2011/0130818; 2011/0238129; 2011/0313500; 2012/0016378; 2012/0046710; 2012/0071949; 2012/0165911; 2012/0197375; 2012/0203316; 2012/0203320; 2012/0203321; 2012/0316615; 2013/0105071; 2011/0005069; 2010/0268298; 2011/0130817; 2011/0130818; 2011/0078900; 2011/0238129; 2011/0313500; 2012/0016378; 2012/0046710; 2012/0165911; 2012/0197375; 2012/0203316; 2012/0203320; and 2012/0203321, all of which are incorporated by reference in their entireties.

(17) Turning to FIG. 1, one embodiment of an electrical stimulation system **10** includes one or more stimulation leads **12** and an implantable pulse generator (IPG) **14**. The system **10** can also include one or more of an external remote control (RC) **16**, a clinician's programmer (CP) **18**, an external trial stimulator (ETS) **20**, or an external charger **22**. The IPG and ETS are examples of control modules for the electrical stimulation system.

(18) The IPG **14** is physically connected, optionally via one or more lead extensions **24**, to the stimulation lead(s) **12**. Each lead carries multiple electrodes **26** arranged in an array. The IPG **14** includes pulse generation circuitry that delivers electrical stimulation energy in the form of, for example, a pulsed electrical waveform (i.e., a temporal series of electrical pulses) to the electrode array **26** in accordance with a set of stimulation parameters. The implantable pulse generator can be implanted into a patient's body, for example, below the patient's clavicle area or within the patient's abdominal cavity or at any other suitable site. The implantable pulse generator **14** can have multiple stimulation channels which may be independently programmable to control the magnitude of the current stimulus from each channel. In some embodiments, the implantable pulse generator **14** can have any suitable number of stimulation channels including, but not limited to, 4, 6, 8, 12, 16, 32, or more stimulation channels. The implantable pulse generator **14** can have one, two, three, four, or more connector ports, for receiving the terminals of the leads and/or lead extensions.

(19) The ETS **20** may also be physically connected, optionally via the percutaneous lead extensions **28** and external cable **30**, to the stimulation leads **12**. The ETS **20**, which may have similar pulse generation circuitry as the IPG **14**, also delivers electrical stimulation energy in the form of, for example, a pulsed electrical waveform to the electrode array **26** in accordance with a set of stimulation parameters. One difference between the ETS **20** and the IPG **14** is that the ETS **20** is often a non-implantable device that is used on a trial basis after the neurostimulation leads **12** have been implanted and prior to implantation of the IPG **14**, to test the responsiveness of the stimulation that is to be provided. Any functions described herein with respect to the IPG **14** can likewise be performed with respect to the ETS **20**.

(20) The RC **16** may be used to telemetrically communicate with or control the IPG **14** or ETS **20** via a uni- or bi-directional wireless communications link **32**. Once the IPG **14** and neurostimulation leads **12** are implanted, the RC **16** may be used to telemetrically communicate with or control the IPG **14** via a uni- or bi-directional communications link **34**. Such communication or control allows the IPG **14**, for example, to be turned on or off and to be programmed with different stimulation parameter sets. The IPG **14** may also be operated to modify the programmed stimulation parameters to actively control the characteristics of the electrical stimulation energy output by the IPG **14**. In at least some embodiments, the CP **18** (or RC **16** or other programming device) allows a user, such as a clinician, the ability to program stimulation parameters for the IPG **14** and ETS **20** in the operating room and in follow-up sessions. Alternately, or additionally, in at least some

embodiments, stimulation parameters can be programmed via wireless communications (e.g., Bluetooth) between the RC **16** (or other external device such as a hand-held electronic device like a mobile phone, tablet, or the like) and the IPG **14**.

(21) The CP **18** may perform this function by indirectly communicating with the IPG **14** or ETS **20**, through the RC **16**, via a wireless communications link **36**. Alternatively, the CP **18** may directly communicate with the IPG **14** or ETS **20** via a wireless communications link (not shown). In at least some embodiments, the stimulation parameters provided by the CP **18** are also used to program the RC **16**, so that the stimulation parameters can be subsequently modified by operation of the RC **16** in a stand-alone mode (i.e., without the assistance of the CP **18**). The CP **18** or RC **16** can be any suitable device including, but not limited to, a computer or other computing device, laptop, mobile device (for example, a mobile phone or tablet), or the like or any combination thereof. The CP **18** or RC **16** can include software applications for interacting with the IPG **14** or ETS **20** and for programming the IPG **14** or ETS **20**.

(22) Additional examples of the RC **16**, CP **18**, ETS **20**, and external charger **22** can be found in the references cited herein as well as U.S. Pat. Nos. 6,895,280; 6,181,969; 6,516,227; 6,609,029; 6,609,032; 6,741,892; 7,949,395; 7,244,150; 7,672,734; and 7,761,165; 7,974,706; 8,175,710; 8,224,450; and 8,364,278; and U.S. Patent Application Publication No. 2007/0150036, all of which are incorporated herein by reference in their entireties.

(23) FIG. **2** is a schematic overview of one embodiment of components of an electrical stimulation system **200** including an electronic subassembly **210** disposed within an IPG **14** (FIG. **1**). It will be understood that the electrical stimulation system can include more, fewer, or different components and can have a variety of different configurations including those configurations disclosed in the stimulator references cited herein.

(24) The IPG **14** (FIG. **1**) can include, for example, a power source **212**, antenna **218**, receiver **202**, processor **204**, and memory **205**. Some of the components (for example, power source **212**, antenna **218**, receiver **202**, processor **204**, and memory **205**) of the electrical stimulation system can be positioned on one or more circuit boards or similar carriers within a sealed housing of the IPG **14** (FIG. **1**), if desired. Unless indicated otherwise, the term “processor” refers to both embodiments with a single processor and embodiments with multiple processors.

(25) An external device, such as a CP or RC **206**, can include a processor **207**, memory **208**, an antenna **217**, and a user interface **219**. The user interface **219** can include, but is not limited to, a display screen on which a digital user interface can be displayed and any suitable user input device, such as a keyboard, touchscreen, mouse, track ball, or the like or any combination thereof.

(26) Any power source **212** can be used including, for example, a battery such as a primary battery or a rechargeable battery. Examples of other power sources include super capacitors, nuclear or atomic batteries, mechanical resonators, infrared collectors, thermally-powered energy sources, flexural powered energy sources, bioenergy power sources, fuel cells, bioelectric cells, osmotic pressure pumps, and the like including the power sources described in U.S. Pat. No. 7,437,193, incorporated herein by reference in its entirety.

(27) As another alternative, power can be supplied by an external power source through inductive coupling via the antenna **218** or a secondary antenna. The external power source can be in a device that is mounted on the skin of the user or in a unit that is provided near the user on a permanent or periodic basis.

(28) If the power source **212** is a rechargeable battery, the battery may be recharged using the antenna **218**, if desired. Power can be provided to the battery for recharging by inductively coupling the battery through the antenna to a recharging unit **216** external to the user. Examples of such arrangements can be found in the references identified above.

(29) In one embodiment, electrical current is emitted by the electrodes **26** on the lead body to stimulate nerve fibers, muscle fibers, or other body tissues near the electrical stimulation system. A processor **204** is generally included to control the timing and electrical characteristics of the

electrical stimulation system. For example, the processor **204** can, if desired, control one or more of the timing, frequency, amplitude, width, and waveform of the pulses. In addition, the processor **204** can select which electrodes can be used to provide stimulation, if desired. In some embodiments, the processor **204** may select which electrode(s) are cathodes and which electrode(s) are anodes. In some embodiments, the processor **204** may be used to identify which electrodes provide the most useful stimulation of the desired tissue. Instructions for the processor **204** can be stored on the memory **205**. Instructions for the processor **207** can be stored on the memory **208**.

(30) Any processor **204** can be used for the IPG and can be as simple as an electronic device that, for example, produces pulses at a regular interval or the processor can be capable of receiving and interpreting instructions from the CP/RC **206** (such as CP **18** or RC **16** of FIG. **1**) that, for example, allows modification of pulse characteristics. In the illustrated embodiment, the processor **204** is coupled to a receiver **202** which, in turn, is coupled to the antenna **218**. This allows the processor **204** to receive instructions from an external source to, for example, direct the pulse characteristics and the selection of electrodes, if desired. Any suitable processor **207** can be used for the CP/RC **206**.

(31) Any suitable memory **205**, **208** can be used including computer-readable storage media may include, but is not limited to, volatile, nonvolatile, non-transitory, removable, and non-removable media implemented in any method or technology for storage of information, such as computer readable instructions, data structures, program modules, or other data. Examples of computer-readable storage media include, but are not limited to, RAM, ROM, EEPROM, flash memory, or other memory technology/#CD-ROM, digital versatile disks (“DVD”) or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to store the desired information and which can be accessed by a processor.

(32) In one embodiment, the antenna **218** is capable of receiving signals (e.g., RF signals) from an antenna **217** of a CP/RC **206** (see, CP **18** or RC **16** of FIG. **1**) which is programmed or otherwise operated by a user. The signals sent to the processor **204** via the antenna **218** and receiver **202** can be used to modify or otherwise direct the operation of the electrical stimulation system. For example, the signals may be used to modify the pulses of the electrical stimulation system such as modifying one or more of pulse width, pulse frequency, pulse waveform, and pulse amplitude. The signals may also direct the electrical stimulation system **200** to cease operation, to start operation, to start signal acquisition, to stop signal acquisition, to start charging the battery, or to stop charging the battery. In other embodiments, the stimulation system does not include an antenna **218** or receiver **202** and the processor **204** operates as programmed.

(33) Optionally, the electrical stimulation system **200** may include a transmitter (not shown) coupled to the processor **204** and the antenna **218** for transmitting signals back to the CP/RC **206** or another unit capable of receiving the signals. For example, the electrical stimulation system **200** may transmit signals indicating whether the electrical stimulation system **200** is operating properly or not or indicating when the battery needs to be charged or the level of charge remaining in the battery. The processor **204** may also be capable of transmitting information about the pulse characteristics so that a user or clinician can determine or verify the characteristics.

(34) Transmission of signals can occur using any suitable method, technique, or platform including, but not limited to, inductive transmission, radiofrequency transmission, Bluetooth™, Wi-Fi, cellular transmission, near field transmission, infrared transmission, or the like or any combination thereof. In addition, the IPG **14** can be wirelessly coupled to the RC **16** or CP **18** using any suitable arrangement include direct transmission or transmission through a network, such as a local area network, wide area network, the Internet, or the like or any combination thereof. The CP **18** or RC **16** may also be capable of coupling to, and sending data or other information to, a network **220**, such as a local area network, wide area network, the Internet, or the like or any combination thereof.

(35) At least some of the stimulation electrodes can take the form of segmented electrodes that extend only partially around the perimeter (for example, the circumference) of the lead. These segmented electrodes can be provided in sets of electrodes, with each set having electrodes circumferentially distributed about the lead at a particular longitudinal position.

(36) In FIGS. 3A, 3B, and 3D the electrodes are shown as including both ring electrodes **120** and segmented electrodes **122**. In some embodiments, the electrodes are all segmented electrode **122**, as illustrated in FIGS. 3C and 3E. The segmented electrodes **122** of FIG. 3A are in sets of three, where the three segmented electrodes of a particular set are electrically isolated from one another and are circumferentially offset along the lead **12**. Any suitable number of segmented electrodes can be formed into a set including, for example, two, three, four, or more segmented electrodes. The lead **12** of FIG. 3A has thirty segmented electrodes **122** (ten sets of three electrodes each) and two ring electrodes **120** for a total of 32 electrodes.

(37) Segmented electrodes can be used to direct stimulus current to one side, or even a portion of one side, of the lead. When segmented electrodes are used in conjunction with an implantable pulse generator that delivers current stimulus, current steering can be achieved to deliver the stimulus more precisely to a position around an axis of the lead (i.e., radial positioning around the axis of the lead). Segmented electrodes may provide for superior current steering than ring electrodes because target structures in deep brain stimulation are not typically symmetric about the axis of the distal electrode array. Instead, a target may be located on one side of a plane running through the axis of the lead. Through the use of a segmented electrode array, current steering can be performed not only along a length of the lead but also around a perimeter of the lead. This provides precise three-dimensional targeting and delivery of the current stimulus to neural target tissue, while potentially avoiding stimulation of other tissue.

(38) FIG. 3A illustrates a 32-electrode lead **12** with a lead body **106** and two ring electrodes **120** proximal to thirty segmented electrodes **122** arranged in ten sets of three segmented electrodes each. In the illustrated embodiments, the ring electrodes **120** are proximal to the segmented electrodes **122**. In other embodiments, the ring electrodes **120** can be proximal to, or distal to, or any combination thereof.

(39) Any number of segmented electrodes **122** may be disposed on the lead body including, for example, one, two, three, four, five, six, seven, eight, nine, ten, eleven, twelve, thirteen, fourteen, fifteen, sixteen, twenty, twenty-four, twenty-eight, thirty, thirty-two, or more segmented electrodes **122**. It will be understood that any number of segmented electrodes **122** may be disposed along the length of the lead body. A segmented electrode **122** typically extends only 75%, 67%, 60%, 50%, 40%, 33%, 25%, 20%, 17%, 15%, or less around the circumference of the lead.

(40) The segmented electrodes **122** may be grouped into sets of segmented electrodes, where each set is disposed around a circumference of the lead **12** at a particular longitudinal portion of the lead **12**. The lead **12** may have any number of segmented electrodes **122** in a given set of segmented electrodes. The lead **12** may have one, two, three, four, five, six, seven, eight, or more segmented electrodes **122** in a given set. The lead **12** may have any number of sets of segmented electrode including, but not limited to, one, two, three, four, five, six, eight, ten, twelve, fifteen, sixteen, twenty, or more sets. The segmented electrodes **122** may be uniform, or vary, in size and shape. In some embodiments, the segmented electrodes **122** are all of the same size, shape, diameter, width or area or any combination thereof. In some embodiments, the segmented electrodes **122** of each circumferential set (or even all segmented electrodes disposed on the lead **12**) may be identical in size and shape.

(41) Each set of segmented electrodes **122** may be disposed around the circumference of the lead body to form a substantially cylindrical shape around the lead body. The spacing between individual electrodes of a given set of the segmented electrodes may be the same, or different from, the spacing between individual electrodes of another set of segmented electrodes on the lead **12**. In at least some embodiments, equal spaces, gaps or cutouts are disposed between each segmented

electrode **122** around the circumference of the lead body. In other embodiments, the spaces, gaps or cutouts between the segmented electrodes **122** may differ in size or shape. In other embodiments, the spaces, gaps, or cutouts between segmented electrodes **122** may be uniform for a particular set of the segmented electrodes **122**, or for all sets of the segmented electrodes **122**. The sets of segmented electrodes **122** may be positioned in irregular or regular intervals along a length of the lead body.

(42) The electrodes of the lead **12** are typically disposed in, or separated by, a non-conductive, biocompatible material of a lead body **106** including, for example, silicone, polyurethane, and the like or combinations thereof. The lead body **106** may be formed in the desired shape by any process including, for example, extruding, molding (including injection molding), casting, and the like. Electrodes and connecting wires can be disposed onto or within a lead body either prior to or subsequent to a molding or casting process. The non-conductive material typically extends from the distal end of the lead body **106** to the proximal end of the lead body **106**.

(43) FIG. 3B to 3E illustrate other embodiments of leads with segmented electrodes **122**. FIG. 3B illustrates a sixteen electrode lead **12** having one ring electrode **120** that is proximal to five sets of three segmented electrodes **122** each. FIG. 3C illustrates a sixteen electrode lead **12** having eight sets of two segmented electrodes **122** each. As illustrated in FIG. 3C, an embodiment of a lead **12** does not necessarily include a ring electrode. FIG. 3D illustrates a sixteen electrode lead **12** having four ring electrodes **120** that are proximal to six sets of two segmented electrodes **122** each. FIG. 3E illustrates a thirty-two electrode lead **12** having sixteen sets of two segmented electrodes **122** each (for clarity of illustration, not all of the electrodes are shown). It will be recognized that any other electrode combination of ring electrodes, segmented electrodes, or both types of electrodes can be used.

(44) When the lead **12** includes both ring electrodes **120** and segmented electrodes **122**, the ring electrodes **120** and the segmented electrodes **122** may be arranged in any suitable configuration. For example, when the lead **12** includes two or more ring electrodes **120** and one or more sets of segmented electrodes **122**, the ring electrodes **120** can flank the one or more sets of segmented electrodes **122**. Alternately, the two or more ring electrodes **120** can be disposed proximal to the one or more sets of segmented electrodes **122** or the two or more ring electrodes **120** can be disposed distal to the one or more sets of segmented electrodes **122**.

(45) The electrodes **120**, **122** may have any suitable longitudinal length including, but not limited to, 2, 3, 4, 4.5, 5, or 6 mm. The longitudinal spacing between adjacent electrodes **120**, **122** may be any suitable amount including, but not limited to, 1, 2, or 3 mm, where the spacing is defined as the distance between the nearest edges of two adjacent electrodes. In some embodiments, the spacing is uniform between longitudinally adjacent electrodes along the length of the lead. In other embodiments, the spacing between longitudinally adjacent electrodes may be different or non-uniform along the length of the lead.

(46) Examples of leads with segmented electrodes include U.S. Patent Application Publications Nos. 2010/0268298; 2011/0005069; 2011/0078900; 2011/0130803; 2011/0130816; 2011/0130817; 2011/0130818; 2011/0078900; 2011/0238129; 2011/0313500; 2012/0016378; 2012/0046710; 2012/0071949; 2012/0165911; 2012/0197375; 2012/0203316; 2012/0203320; 2012/0203321; 2013/0197602; 2013/0261684; 2013/0325091; 2013/0317587; 2014/0039587; 2014/0353001; 2014/0358209; 2014/0358210; 2015/0018915; 2015/0021817; 2015/0045864; 2015/0021817; 2015/0066120; 2013/0197424; 2015/0151113; 2014/0358207; and U.S. Pat. No. 8,483,237, all of which are incorporated herein by reference in their entireties. A lead may also include a tip electrode and examples of leads with tip electrodes include at least some of the previously cited references, as well as U.S. Patent Application Publications Nos. 2014/0296953 and 2014/0343647, all of which are incorporated herein by reference in their entireties. A lead with segmented electrodes may be a directional lead that can provide stimulation in a particular direction using the segmented electrodes.

(47) Although electrical stimulation of tissue has demonstrated effectiveness for many medical conditions, there can be side-effects to electrical stimulation. Multi-electrode leads may enable greater selectivity of tissue to be stimulated which may provide improved therapy, but there can be a challenge in determining which electrodes and stimulation parameters provide therapeutic benefit while reducing or eliminating side effects.

(48) In at least some instances, a treating physician may wish to tailor the stimulation parameters (such as which one or more of the stimulating electrodes to use, the stimulation pulse amplitude (such as current or voltage amplitude depending on the stimulator being used,) the stimulation pulse width, the stimulation frequency, the duty cycle, the stimulation phase, or the like or any combination thereof) for a particular patient. Conventionally, the clinician or programmer tries different selections of electrodes, stimulation amplitude, and other parameters and then determines therapy effectiveness or side effects using visual observation, sensor observation, patient feedback, or the like. This can be a lengthy and time-consuming process.

(49) In contrast to the conventional programming methods, as described herein, an automated (or at least partially automated) programming system or method can include the automated sequential testing of different sets of stimulation parameters and evaluation of each set of stimulation parameters using a sensor or sensed responses to the stimulation. The automated programming system or method can select one or more of the tested sets of stimulation parameters for therapy. In at least some embodiments, the automation can then be fine-tuned (or otherwise altered) by manual adjustment of the stimulation parameter by a clinician or other programmer.

(50) FIG. 4 illustrates one embodiment of a method of determining stimulation parameters for therapy. In step **402**, a user, such as a clinician or other programmer, sets limits on one or more stimulation parameters. Examples of stimulation parameters that can be limited include, but are not limited to, amplitude, electrode selection, stimulation direction, pulse width, pulse frequency, or the like or any combination thereof. For example, the user may place a lower limit on the amplitude because lower amplitudes are unlikely to produce satisfactory effects or the user may place an upper limit because higher amplitudes are likely to produce side effects. The user may place limits on which electrodes are selected on the lead (for example, electrodes 2 to 5 in an eight-electrode array) so that only electrodes that are near the target region are selected. For leads with segmented electrodes, the user may limit the direction of the stimulation so that only certain segmented electrodes (within those limits on direction) will be tested. This may reflect knowledge of the position and orientation of the lead in the tissue and the desired target region for stimulation.

(51) Steps **404** to **412** form a loop in which a processor automatically tests multiple sets of stimulation parameters, constrained by the limits set in step **402**, and determines the effect(s) of the test stimulations. The effect(s) can be therapeutic effect(s) or side effect(s) or any combination thereof. In at least some embodiments, the loop is closed and occurs automatically. In at least some embodiments, the loop may occur with no manual input. In at least some embodiments, the loop may occur with manual input limited to input from the clinician, programmer, or patient regarding observed therapeutic or side effects. In at least some embodiments, the loop can be halted manually by the user, clinician, programmer, patient, or any combination thereof. This manual control can be used, for example, to halt the process if there are safety or patient comfort concerns or issues. In at least some embodiments, the processor may also include one or more thresholds (for example, threshold values of side effects) that, when met or exceeded, may halt the loop as well. In at least some embodiments, halting the loop based on one or more thresholds can result from the sensing of therapeutic or side effects described below.

(52) In step **404**, a set of stimulation parameters is selected. In at least some embodiments, one or more (or even all of the) sets of stimulation parameters that are to be tested are determined prior to the start of the initial loop of steps **404** to **412**. In at least some embodiments, additional sets of stimulation parameters may be determined during, or as a result of, the loop of steps **404** to **412** based, for example, on the effect(s) sensed during the testing. Non-limiting examples of procedures

for determining sets of parameters based on feedback or sensed effects are described in U.S. Pat. No. 10,603,498, which is incorporated herein by reference in its entirety.

(53) In at least some embodiments, the testing of the sets of stimulation parameters can be performed according to a predefined routine or order. For example, the predefined routine or order for testing different electrode selections may start at a proximal position and move distally for each subsequent set or start at a distal position and move proximally for each subsequent set. A predefined routine or order for testing different amplitudes may start at a low or intermediate amplitude and increase for each subsequent set.

(54) In other embodiments, the testing of the sets of stimulation parameters can be performed in any other suitable order including random, pseudo-random, partially random, or based on any other selection method or criterion/criteria.

(55) In at least some embodiments, the testing of the sets of stimulation parameters can be performed using coarse testing of sets that differ in one parameter, followed by fine-tuning for that parameter, and then proceeding to do the same for one or more additional parameters. Examples of methods for testing the sets of stimulation parameters are presented below in the embodiments illustrated in FIG. 5.

(56) In step **406**, the patient is stimulated using the selected set of stimulation parameters. In step **408**, one or more effects of the stimulation are sensed. Any suitable therapeutic effects, side effects, response effects, or any combination thereof can be sensed. A response effect can be any effect that is a result of the stimulation and can include, but are not limited to, therapeutic (e.g., beneficial) effects or side (e.g., detrimental) effects. Such response effects can be, for example, neural response such as an evoked compound action potential (ECAP), an evoked resonant neural activity (ERNA), or a local field potential (LFP). These response effects are not necessarily therapeutic or side effects.

(57) In at least some embodiments, one or more effects are sensed by a sensor external to the IPG, by a sensor coupled to the IPG, or by one or more of the electrodes of the leads (acting as sensor(s)), or the like or any combination thereof. In at least some embodiments, the sensor or electrodes are used to obtain measurements of the effect(s). In at least some embodiments, the IPG, CP, RC, ETS, or other device are used to evaluate the measurements from the sensor or electrodes. Examples of methods for sensing the effect(s) are presented below including the embodiments illustrated in FIG. 6, as well as in U.S. Pat. Nos. 10,357,657; 10,369,364; and 10,716,505 and U.S. Patent Application Publications Nos. 2014/0243926; 2014/0276707; 2014/0277282; 2020/0376263; 2020/0398057; and 2021/0023374, all of which are incorporated herein by reference in their entireties.

(58) In at least some embodiments, the valuation of the measurements from the sensor or electrodes can include a comparative evaluation of the therapeutic effect(s) and side effect(s). In at least some embodiments, an intensity or benefits scale can be used to rate the therapeutic effect(s) and an intensity or detriment scale can be used to rate the side effect(s). In at least some embodiments, these two scales can be selected for comparison. For example, in at least some embodiments, a therapeutic score may be determined by subtracting the rating for the side effect(s) from the rating for the therapeutic effect(s). The higher the therapeutic score, the likelihood that the corresponding stimulation is beneficial. Other algorithms or methods of comparing the therapeutic effect(s) to the side effect(s) can be used.

(59) In step **410**, a collection of the effects and sets of stimulation parameters are updated with the current set of stimulation parameters and the sensed effect(s). The collection can be in any suitable form including, but not limited to, a database, a clinical effects map or graph, or the like or any combination thereof. FIG. 7 illustrates one embodiment of a clinical effects map. Additional information regarding clinical effects maps can be found in, for example, U.S. Pat. Nos. 9,227,074; 9,248,296; 9,358,398; 9,474,903; 10,603,498; and 10,071,249 and U.S. Patent Application Publication No. 2018/0264278, all of which are incorporated herein in their entireties.

(60) FIG. 7 illustrates one embodiment of a user interface **770** that includes a clinical effects map **772** that maps the electrode position along the y-axis and stimulation amplitude (in mA) along the x-axis. Examples of three-dimensional clinical effects maps and clinical effects maps using cylindrical coordinates (for example, where rho represent amplitude, phi represents a rotational angle, and z represents electrode position) are presented in the references cited above. Positions on the clinical effects map **772** correspond to stimulations, each associated with activation of an actual or virtual electrode at a particular stimulation amplitude. The virtual electrodes can correspond to stimulation using a combination of actual electrodes. For example, a position between electrode 1 and electrode 2 can correspond to stimulation using a combination of electrodes 1 and 2 (for example, 50% of the stimulation amplitude on electrode 1 and 50% of the stimulation amplitude on electrode 2 to provide a virtual electrode with a midpoint between electrodes 1 and 2).

(61) In the illustrated embodiment, the markings **774** on the clinical effects map **772** correspond to actual stimulation events. In at least some embodiments, the markings **774** may also provide information about the effect(s) produced by the stimulation. In the illustrated embodiment of FIG. 7, the markings can have a circle **774a** that corresponds to therapeutic effect(s) with variation in color or intensity indicating the strength of the therapeutic effect(s). The markings can have a ring **774b** that corresponds to side effect(s) with variation in color or intensity indicating the strength of the side effect(s). In the illustrated embodiment, the user interface **770** includes an input section **776** with controls **778** for inputting specific types of therapeutic or side effects and corresponding intensities.

(62) The clinical effects map **772** may also include a feature that indicates potential target regions **780a** and potential avoidance regions **780b** that are determined based on the preceding stimulation instances. For example, the potential target regions **780a** can be indicated on the clinical effects map **772** as a particular background color (for example, green) and correspond to regions near stimulation instances in which the therapeutic effect(s) are greater than the side effect(s) (and, optionally, where the side effect(s) are below a threshold level). The potential avoidance regions **780b** can be indicated on the clinical effects map **772** as a different background color (for example, red) and correspond to regions where the trendlines from the preceding stimulation instances indicate that the side effect(s) are likely greater than the therapeutic effect(s) or where the side effects are likely to meet or exceed a threshold level. The potential avoidance region **780b** may also include the region out the limits set by the user in step **402**, as described above.

(63) In step **412**, the processor determines if there is another set of stimulation parameters to test. If yes, then the method returns to step **404**. If no, then the method continues.

(64) In step **414**, the processor selects one of the sets of stimulation parameters based on the testing. In at least some embodiments, the set of stimulation parameters with the best therapeutic response is selected. In at least some embodiments, the processor will select the set of stimulation parameters based on both the therapeutic response and the presence or severity of side effects. In at least some embodiments, the ratings of the therapeutic effect(s), side effect(s), or any combination thereof can be used to select the set of stimulation parameters. In at least some embodiments, the therapeutic scores (e.g., the different between the rating for the therapeutic effect(s) and the rating for the side effect(s)) for the sets of stimulation parameters can be used to select the set of stimulation parameters.

(65) In optional step **416**, a clinician or programmer can fine-tune the stimulation parameters by altering or modifying the set of stimulation parameters selected by the processor in step **414**. In at least some embodiments, the processor may require that the clinician or programmer authorize or accept the selected set of stimulation parameters. In step **418**, the IPG is programmed using the selected set of stimulation parameters or the set of stimulation parameters after modification or alteration by the clinician or programmer in step **416**.

(66) In at least some embodiments, steps **404** to **412** can be performed while the patient is asleep when the testing is performed without patient or clinician feedback. In at least some embodiments,

the method can result in personalized therapy for the patient or can provide programming that is tailored to the patient's physiological response.

(67) In at least some embodiments, the method can be performed periodically and can be used to track disease progression over time. In at least some embodiments, the method can be performed periodically to adjust stimulation parameters or to improve closed-loop device operation.

(68) One embodiment of a method for selection of a set of stimulation parameters is illustrated in FIG. 5. In step 502, a number (for example, two, three, four, five, six, eight, ten, twelve, or more) of different axial positions along the lead are tested. Each of the axial positions can correspond to a single electrode (or a set of segmented electrodes at the same axial position along the lead) or to multiple electrodes (including multiple electrodes at different axial positions along the lead to provide a virtual electrode that is a combination of these electrodes.) As an example, an axial position between electrode 1 (or a set of segmented electrodes at axial position 1 along the lead) and electrode 2 (or a set of segmented electrodes at axial position 2 along the lead) can correspond to stimulation using a combination of electrodes 1 and 2. For example, an axial position of 1.5 may correspond to 50% of the stimulation amplitude on electrode 1 and 50% of the stimulation amplitude on electrode 2 and an axial position of 1.25 may correspond to 75% of the stimulation amplitude on electrode 1 and 25% of the stimulation amplitude on electrode 2.

(69) In at least some embodiments, the different axial positions are selected to be uniformly spread along the portion of the lead that is permitted for use. In at least some embodiments, when a particular axial position produces side effects over a threshold level or produces therapeutic effects below a particular threshold, other axial positions near, distal, or proximal to the particular axial position may be precluded from testing. The processor may determine that the precluded axial positions may produce side effects or not produce substantial therapeutic effects.

(70) The processor selects one or more of these initial axial positions for proceeding. For example, the processor may use a comparative evaluation of the therapeutic effect(s) and side effect(s). In at least some embodiments, an intensity or benefits scale can be used to rate the therapeutic effect(s) and an intensity or detriment scale can be used to rate the side effect(s). In at least some embodiments, these two scales can be selected for comparison. For example, in at least some embodiments, a therapeutic score may be determined by subtracting the rating for the side effect(s) from the rating for the therapeutic effect(s). The higher the therapeutic score, the likelihood that the corresponding stimulation is beneficial. Other algorithms or methods of comparing the therapeutic effect(s) to the side effect(s) can be used.

(71) In optional step 504, the processor may fine-tune the axial position by testing additional axial positions near the selected initial axial positions. For example, if the axial position corresponding to electrode 3 (or a set of segmented electrodes at axial position 3 along the lead) is selected, then fine-tuning may include testing sets of stimulation parameters with axial positions in the range of 2 to 4 (e.g., 2.25, 2.5, 2.75, 3.25, 3.5, and 3.75) which corresponds axial positions within one axial electrode spacing of axial position 3. In at least some embodiments, the fine-tuning includes testing axial positions within one, two, three, or four or more axial electrode spacings from the selected initial axial position. The processor then selects one or more these fine-tuned axial positions (or the initial axial positions) for proceeding using any suitable method including those described above.

(72) In optional step 506, when the selected initial or fine-tuned axial positions include segmented electrodes, the processor can select two or more different rotational directions at the selected axial positions. Each of the rotational directions can correspond to a single segmented electrode or to multiple segmented electrodes (including multiple segmented electrodes at different axial positions along the lead to provide a virtual electrode that is a combination of these segmented electrodes.) As an example, a rotational direction between segmented electrode 1 and segmented electrode 2 can correspond to stimulation using a combination of segmented electrodes 1 and 2. For example; a rotational direction of 1.5 may correspond to 50% of the stimulation amplitude on segmented electrode 1 and 50% of the stimulation amplitude on segmented electrode 2.

(73) The processor selects one or more of these initial rotational directions for proceeding. In optional step **508**, the processor can also fine-tune the rotational direction(s) by testing additional rotational directions near the selected initial rotational direction. For example, if the rotational direction corresponding to electrode 2 is selected, then fine-tuning may include testing sets of stimulation parameters with rotational directions in the range of 1 to 3 (e.g., 1.25, 1.5, 1.75, 2.25, 2.5, and 2.75). In at least some embodiments, the fine-tuning includes testing rotational directions within 120, 100, 90, 60, 45, or fewer degrees from the selected initial rotational direction. One of the fine-tuned rotational directions (or initial rotational direction(s)) can be selected for proceeding using any suitable method including those described above.

(74) In step **510**, one or more additional stimulation parameters can be tested using the selected axial position and optional rotational direction. For example, different stimulation amplitudes can be tested. Other stimulation parameters can include pulse duration, pulse frequency, or the like or any combination thereof. For each of these one or more additional stimulation parameters, a value can be selected based on the testing using any suitable method including those described above.

(75) One embodiment of a method of sensing a stimulation effect is provided in FIG. **6**. In step **602**, after stimulating with a set of stimulation parameters, the sensor makes a measurement of one or more effect(s) of the stimulation including, but not limited to, therapeutic effects, side effects, response effects, or the like or any combination thereof. The stimulation effect can be, for example, an evoked compound action potential (ECAP), an evoked resonant neural activity (ERNA), a local field potential (LFP), or the like or any combination thereof. In some embodiments, the stimulation effect can be an effect of the disease or disorder such as rigidity, tremor, impaired or unwanted movement, synchronized signals in the brain or elsewhere, seizures, or the like or any combination thereof. The sensor can be, for example, a sensor external to the IPG, by a sensor coupled to the IPG, or by one or more of the electrodes of the leads (acting as sensor(s)), or the like or any combination thereof. In at least some embodiments, there may be at least some digital signal processing of the measurement in step **602**, such as, for example, finite impulse response (FIR) filtering or recursively averaging epochs of data.

(76) In step **604**, one or more features of the measurements are extracted. For example, in at least some embodiments, the measurements can be signal that is recorded over time and one or more features of the signal can be extracted. FIG. **8** illustrates one embodiment of a signal **880** measured using a sensor (such as an external or internal sensor or one or more of the electrodes of the lead.) In FIG. **8**, the signal **880** is presented as a graph of current (or voltage) amplitude (y-axis) versus time (x-axis). Examples of the measurement of such signals **880** can be found at, for example, U.S. Pat. Nos. 10,357,657; 10,369,364; and 10,716,505 and U.S. Patent Application Publications Nos. 2014/0243926; 2014/0276707; 2014/0277282; 2020/0376263; 2020/0398057; and 2021/0023374, all of which are incorporated herein by reference in their entireties.

(77) Non-limiting examples of features that can be extracted from the signal **880** include, for example, the range of amplitude, value(s) of positive peaks, value(s) of negative peaks, area under the signal curve, length of the signal curve, or the like or any combination thereof. In at least some embodiments, a range of time is selected or predefined for the extraction of the feature(s). For example, the range of time may be a block of 64, 128, 256, or 512 (or any other suitable number) of samples of the measurements made by the sensor.

(78) Signal **880** of FIG. **8** can be used as an example, to illustrate different measurements. For example, a y-range measurement can correspond to $y_{\text{sub.max}} - y_{\text{sub.min}}$ which in the case of signal **880** is $8 - 1 = 7$. An x-range measurement can correspond, for example, to the x-range between $y_{\text{sub.max}}$ (e.g., $x_{\text{sub.y-max}} = 3$ for signal **880**) and $x_{\text{sub.min}}$ (e.g., $x_{\text{sub.y-min}} = 7$ for signal **880**) which in the case of signal **880** is $(7 - 3) = 4$.

(79) Another example is the area under the curve of the signal. This area can be determined using any suitable method including integration techniques or the like. One example uses the algorithm $\text{Area Under Curve} = \sum |y|$ at each value of x which for signal **880** is $6 + 5 + 8 + 6 + 3 + 4 + 1 = 33$.

(80) Yet another example, is the length of the curve of the signal. This length can be determined using any suitable method for determining or estimating a length of a curve. One example uses the algorithm $\text{Curve Length} = \sum |\Delta y| = \sum |y_{\text{sub.i}} - y_{\text{sub.i-1}}|$ which for signal **880** is $2+1+3+2+3+1+3=15$.

(81) The methods, techniques, and systems described herein are presented in the context of an electrical stimulation system, but it will be recognized that these methods, techniques, and systems can be used with an optical stimulation system or an electrical/optical stimulation system. Examples of optical stimulation systems or electrical/optical stimulation systems are found in U.S. Patent Application Publications Nos. 2013/0317572; 2013/0317573; 2017/0259078; 2017/0225007; 2018/0110971;

(82) 2018/0369606; 2018/0369607; 2019/0209849; 2019/0209834; 2020/0094047; and 2020/0155854 and U.S. patent application Ser. No. 16/883,404, all of which are incorporated herein by reference in their entireties.

(83) It will be understood that each block of the flowchart illustration, and combinations of blocks in the flowchart illustration and methods disclosed herein, can be implemented by computer program instructions. These program instructions may be provided to a processor to produce a machine or engine, such that the instructions, which execute on the processor, create means for implementing the actions specified in the flowchart block or blocks or engine disclosed herein. The computer program instructions may be executed by a processor to cause a series of operational steps to be performed by the processor to produce a computer implemented process. The computer program instructions may also cause at least some of the operational steps to be performed in parallel. Moreover, some of the steps may also be performed across more than one processor, such as might arise in a multi-processor computing device. In addition, one or more processes may also be performed concurrently with other processes, or even in a different sequence than illustrated without departing from the scope or spirit of the invention.

(84) The computer program instructions can be stored on any suitable computer-readable medium including, but not limited to, RAM, ROM, EEPROM, flash memory or other memory technology, CD-ROM, digital versatile disks (“DVD”) or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to store the desired information and which can be accessed by a computing device. The computer program instructions can be stored locally or nonlocally (for example, in the Cloud).

(85) The above specification and examples provide a description of the arrangement and use of the invention. Since many embodiments of the invention can be made without departing from the spirit and scope of the invention, the invention also resides in the claims hereinafter appended.

Claims

1. A method for automating selection of stimulation parameters for a stimulation device implanted in a patient, the method comprising: setting, by a user, at least one limit on each of at least three stimulation parameters, each of the at least three stimulation parameters being different and selected from an axial position along an electrical stimulation lead, a rotational direction of stimulation, a stimulation amplitude, a pulse duration, a pulse frequency, or a duty cycle, wherein each of the at least one limit comprises an upper limit, a lower limit, a limit on electrode selection, or a limit on stimulation direction; and performing, automatically using at least one processor, the following actions sequentially and individually for each of the at least three stimulation parameters constrained by the at least one limit, wherein the actions are fully performed with respect to one of the at least three stimulation parameters before performing the actions for another one of the at least three stimulation parameters: selecting a one of the at least three stimulation parameters, stimulating tissue of the patient, by the stimulation device, using different first values for the selected one of the at least three stimulation parameters constrained by the at least one limit for the one of the at least three stimulation parameters, for each of the different first values, sensing one or

more first effects arising in response to the corresponding stimulation including sensing a signal from the stimulated tissue, wherein the signal is an evoked compound action potential (ECAP), evoked resonant neural activity (ERNA), or local field potential (LFP), wherein at least one of the one or more first effects is a feature of the signal over a selected range of time, wherein the feature comprises at least one of a range of amplitude of the signal over the selected range of time, an amount of time between a maximum amplitude of the signal and a minimum amplitude of the signal for the selected range of time, an area under a curve of the signal for the selected range of time, or a length of the curve of the signal for the selected range of time, updating, by the at least one processor and for each of the different first values, a collection of effects and corresponding values of the at least three stimulation parameters with the one or more first effects and the corresponding different first values of the one of the at least three stimulation parameters, selecting, by the processor, a coarse value from the different first values of the one of the at least three stimulation parameters based on the sensed one or more first effects, stimulating tissue of the patient, by the stimulation device, using different second values for the selected one of the at least three stimulation parameters that are within a predefined relative range around the selected coarse value of the one of the at least three stimulation parameters constrained by the at least one limit for the one of the at least three stimulation parameters, for each of the different second values, sensing one or more second effects arising in response to the stimulation including sensing a signal from the stimulated tissue, wherein the signal is an evoked compound action potential (ECAP), evoked resonant neural activity (ERNA), or local field potential (LFP), wherein at least one of the one or more second effects is a feature of the signal over a selected range of time, wherein the feature comprises at least one of a range of amplitude of the signal over the selected range of time, an amount of time between a maximum amplitude of the signal and a minimum amplitude of the signal for the selected range of time, an area under a curve of the signal for the selected range of time, or a length of the curve of the signal for the selected range of time, updating, by the at least one processor and for each of the different second values, a collection of effects and corresponding values of the at least three stimulation parameters with the one or more second effects and the corresponding different second values of the one of the at least three stimulation parameters, and selecting, by the processor, a fine-tuned value from the different second values of the one of the at least three stimulation parameters based on the sensed one or more second effects, wherein, after the fine-tuned value is selected for any one of the at least three stimulation parameters, that fine-tuned value is used in subsequent stimulations when performing the actions for another one of the at least three stimulation parameters.

2. The method of claim 1, further comprising, after selecting the fine-tuned value of the one of the at least three stimulation parameters, further fine-tuning, manually by the user, to obtain a manual fine-tuned value for one of the at least three stimulation parameters.
3. The method of claim 1, further comprising programming the stimulation device with the selected fine-tuned values for each of the at least three stimulation parameters.
4. The method of claim 1, wherein the signal is an evoked compound action potential (ECAP) or an evoked resonant neural activity (ERNA).
5. The method of claim 1, wherein the at least three stimulation parameters comprises a first stimulation parameter corresponding to an axial location along the electrical stimulation lead.
6. The method of claim 5, wherein the predefined relative range is one axial electrode spacing on both sides of the selected coarse value of the axial location.
7. The method of claim 5, wherein the at least three stimulation parameters comprises a second stimulation parameter corresponding to a rotational direction around the electrical stimulation lead.
8. The method of claim 7, wherein the predefined relative range is 120 degrees on both sides of the selected coarse value of the rotational direction.
9. The method of claim 7, wherein the at least three stimulation parameters comprises a third stimulation parameter corresponding to a stimulation amplitude.

10. A system for selection of stimulation parameters for a stimulation device implanted in a patient, the system comprising: at least one processor configured to perform actions, the actions comprising: receiving, from a user, at least one limit for each of at least three stimulation parameters, each of the at least three stimulation parameters being different and selected from an axial position along an electrical stimulation lead, a rotational direction of stimulation, a stimulation amplitude, a pulse duration, a pulse frequency, or a duty cycle, wherein each of the at least one limit comprises an upper limit, a lower limit, a limit on electrode selection, or a limit on stimulation direction; and performing the following actions sequentially and individually for each of the at least three stimulation constrained by the at least one limit, wherein the actions are fully performed with respect to one of the at least three stimulation parameters before performing the actions for another one of the at least three stimulation parameters: selecting a one of the at least three stimulation parameters, directing the stimulation device to stimulate tissue of the patient using different first values for the selected one of the at least three stimulation parameters constrained by the at least one limit for the one of the at least three stimulation parameters, for each of the different first values, sensing one or more first effects arising in response to the corresponding stimulation including sensing a signal from the stimulated tissue, wherein the signal is an evoked compound action potential (ECAP), evoked resonant neural activity (ERNA), or local field potential (LFP), wherein at least one of the one or more effects is a feature of the signal over a selected range of time, wherein the feature comprises at least one of a range of amplitude of the signal over the selected range of time, an amount of time between a maximum amplitude of the signal and a minimum amplitude of the signal for the selected range of time, an area under a curve of the signal for the selected range of time, or a length of the curve of the signal for the selected range of time, updating, for each of the different first values, a collection of the effects and corresponding values of the at least three stimulation parameters with the one or more first effects and the corresponding different first values of the one of the at least three stimulation parameters, selecting a coarse value from the different first values of the one of the at least three stimulation parameters based on the sensed one or more first effects, directing stimulation of tissue of the patient, by the stimulation device, using different second values for the selected one of the at least three stimulation parameters that are within a predefined relative range around the selected coarse value of the one of the at least three stimulation parameters constrained by the at least one limit for the one of the at least three stimulation parameters, for each of the different second values, sensing one or more second effects arising in response to the stimulation including sensing a signal from the stimulated tissue, wherein the signal is an evoked compound action potential (ECAP), evoked resonant neural activity (ERNA), or local field potential (LFP), wherein at least one of the one or more second effects is a feature of the signal over a selected range of time, wherein the feature comprises at least one of a range of amplitude of the signal over the selected range of time, an amount of time between a maximum amplitude of the signal and a minimum amplitude of the signal for the selected range of time, an area under a curve of the signal for the selected range of time, or a length of the curve of the signal for the selected range of time, updating, by the at least one processor and for each of the different second values, a collection of effects and corresponding values of the at least three stimulation parameters with the one or more second effects and the corresponding different second values of the one of the at least three stimulation parameters, and selecting, by the processor, a fine-tuned value from the different second values of the one of the at least three stimulation parameters based on the sensed one or more second effects, wherein, after the fine-tuned value is selected for any one of the at least three stimulation parameters, that fine-tuned value is used in subsequent stimulations when performing the actions for another one of the at least three stimulation parameters.

11. The system of claim 10, further comprising the stimulation device.

12. The system of claim 11, wherein the stimulation device comprises the electrical stimulation lead having electrodes disposed along a distal portion of the electrical stimulation lead.

13. The system of claim 12, wherein sensing one or more first effects comprising sensing the one or more first effects arising in response to the stimulation using at least one of the electrodes of the electrical stimulation lead.

14. The system of claim 10, further comprising a sensor external to the stimulation device, wherein sensing one or more first effects comprising sensing the one or more first effects arising in response to the stimulation using the sensor.

15. The system of claim 10, wherein the actions further comprise programming the stimulation device with the fine-tuned values selected for the at least three stimulation parameters.

16. A non-transitory computer-readable medium having stored thereon instructions for execution by at least one processor, the instructions comprising: a) receiving, from a user, at least one limit for each of at least three stimulation parameters, each of the at least three stimulation parameters being different and selected from an axial position along an electrical stimulation lead, a rotational direction of stimulation, a stimulation amplitude, a pulse duration, a pulse frequency, or a duty cycle, wherein each of the at least one limit comprises an upper limit, a lower limit, a limit on electrode selection, or a limit on stimulation direction; and b) performing the following actions sequentially and individually for each of the at least three stimulation constrained by the at least one limit, wherein the actions are fully performed with respect to one of the at least three stimulation parameters before performing the actions for another one of the at least three stimulation parameters: selecting a one of the at least three stimulation parameters, directing the stimulation device to stimulate tissue of the patient using different first values for the selected one of the at least three stimulation parameters constrained by the at least one limit for the one of the at least three stimulation parameters, for each of the different first values, sensing one or more first effects arising in response to the corresponding stimulation including sensing a signal from the stimulated tissue, wherein the signal is an evoked compound action potential (ECAP), evoked resonant neural activity (ERNA), or local field potential (LFP), wherein at least one of the one or more effects is a feature of the signal over a selected range of time, wherein the feature comprises at least one of a range of amplitude of the signal over the selected range of time, an amount of time between a maximum amplitude of the signal and a minimum amplitude of the signal for the selected range of time, an area under a curve of the signal for the selected range of time, or a length of the curve of the signal for the selected range of time, updating, for each of the different first values, a collection of the effects and corresponding values of the at least three stimulation parameters with the one or more first effects and the corresponding different first values of the one of the at least three stimulation parameters, selecting a coarse value from the different first values of the one of the at least three stimulation parameters based on the sensed one or more first effects, directing stimulation of tissue of the patient, by the stimulation device, using different second values for the selected one of the at least three stimulation parameters that are within a predefined relative range around the selected coarse value of the one of the at least three stimulation parameters constrained by the at least one limit for the one of the at least three stimulation parameters, for each of the different second values, sensing one or more second effects arising in response to the stimulation including sensing a signal from the stimulated tissue, wherein the signal is an evoked compound action potential (ECAP), evoked resonant neural activity (ERNA), or local field potential (LFP), wherein at least one of the one or more second effects is a feature of the signal over a selected range of time, wherein the feature comprises at least one of a range of amplitude of the signal over the selected range of time, an amount of time between a maximum amplitude of the signal and a minimum amplitude of the signal for the selected range of time, an area under a curve of the signal for the selected range of time, or a length of the curve of the signal for the selected range of time, updating, by the at least one processor and for each of the different second values, a collection of effects and corresponding values of the at least three stimulation parameters with the one or more second effects and the corresponding different second values of the one of the at least three stimulation parameters, and selecting, by the processor, a fine-tuned

value from the different second values of the one of the at least three stimulation parameters based on the sensed one or more second effects, wherein, after the fine-tuned value is selected for any one of the at least three stimulation parameters, that fine-tuned value is used in subsequent stimulations when performing the actions for another one of the at least three stimulation parameters.

17. The non-transitory computer-readable medium of claim 16, wherein the instructions further comprise, after selecting the fine-tuned value of the one of the at least three stimulation parameters, further fine-tuning, manually by the user, to obtain a manual fine-tuned value for one of the at least three stimulation parameters.

18. The non-transitory computer-readable medium of claim 16, wherein the instructions further comprising repeating the instruction b) periodically to track disease progression over time.
